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Two Blind Spots in Backtest Validation: From a Manufactured Dollar Index Anomaly to a Corrected Null Result

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Abstract

Standard backtest validation corrects for the number of strategies tried, but it takes two things as given: the input data and the declared search space. We study what happens when each assumption fails, using the U.S. Dollar Index and a single codebase. On a widely used free data source, an overnight/daytime decomposition produces an overnight Sharpe ratio of -1.43 and a combined strategy with a Sharpe ratio of 1.06, and overfitting diagnostics rate the result as robust (probability of backtest overfitting 0.02). On an independently sourced futures series covering the same sessions, the overnight leg is indistinguishable from zero (Sharpe $+0.25$, $p = 0.33$). The evidence is consistent with a persistent downward bias in the free series' recorded opening prices rather than with a market effect. A binomial screen on where the open sits within the session range flags the free series ($p = 3.7 \times 10^{-13}$ if sessions are independent, and 7×10^{-7} to 0.013 with Newey–West corrections for the clustering of the flagged sessions). In the simulated settings we examined, the screen keeps its nominal size under symmetric microstructure noise and detects directional opening-price errors, but it cannot detect symmetric ones. Separately, a regime-conditioned currency strategy (Sharpe 0.72 on six reported configurations) does not survive correction once its full 216-configuration search is declared. A 1,312-configuration search on futures data finds no configuration above the best Sharpe ratio the same search attains on block-bootstrapped returns (mean 0.74 over 2,000 replications; the observed best, 0.65, sits at the 23rd percentile of that null distribution). We propose a four-stage reporting protocol that separates data validation, strategy specification, multiplicity correction and economic interpretation.

Keywords: backtest overfitting; data quality; deflated Sharpe ratio; foreign exchange; multiple testing; overnight returns; reproducibility; trading strategy evaluation; U.S. Dollar Index

JEL Classification: C12; C15; C52; F31; G14

1. Introduction

The standard defense of a quantitative backtest addresses one failure: that too many strategies were tried. The analyst enumerates the strategies tested and applies a correction for the number of trials, such as White's Reality Check [1], Hansen's test of superior predictive ability [2], the stepwise procedure of Romano and Wolf [3], the deflated Sharpe ratio of Bailey and López de Prado [4], the probability of backtest overfitting of Bailey et al. [5], or the haircut Sharpe ratio of Harvey and Liu [6]. These tools are well-developed and widely used, including for sub-period decompositions of the type used in this paper [7].

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Each of these tools takes two things as given: the input data, and the set of strategies the analyst declares. Two failures, therefore, fall outside what they can see. If the input series is wrong, the correction is applied faithfully to a number that does not describe any market. If the declared strategy set is smaller than the set actually explored, the correction is applied to the wrong denominator. Both problems are known separately, the first from the literature on vendor data quality [8] and the second from the literature on data snooping [9–11]. They are rarely studied together, and the interaction between a data defect and the overfitting diagnostics has, to our knowledge, not been documented on real data.

Research questions. The paper addresses three questions.

RQ1 (data). Can a defect in recorded opening prices produce an apparent trading anomaly that the standard multiple-testing and overfitting diagnostics rate as robust, and can such a defect be screened for using only the series itself?

RQ2 (search space). How does inference on a published-style strategy change once the full set of configurations that were available to the search is declared, and how should implementation errors be separated from search decisions?

RQ3 (benchmark and protocol). What does a correctly specified, fully declared search on validated data find in this market, measured against what the same search finds on returns with no exploitable time-series structure, and what reporting protocol follows from the answers to RQ1 and RQ2?

We answer these questions on one instrument, the U.S. Dollar Index (DXY), with one codebase. For RQ1 we run a complete overnight/daytime decomposition of the DXY on a widely used free data source (Sections 4–8), then compare it with an independently sourced futures series for the same sessions (Section 9). For RQ2 we re-examine a regime-conditioned currency strategy whose reported configurations were a small subset of its search space (Section 10). For RQ3 we run a 1,312-configuration search built from published specifications and benchmark it against a bootstrap null (Sections 11 and 12).

Contributions. The paper makes one methodological contribution, supported by two empirical case studies and one practical proposal.

Core methodological contribution. We show on real data that overfitting diagnostics can rank a result produced by a persistent data defect above the same analysis on an independently sourced comparator. On the free series, the probability of backtest overfitting is 0.016 to 0.028 and Hansen's test rejects at $p = 0.013$; on the futures comparator, the same diagnostics return 0.765 to 0.779 and $p = 1.000$. A defect present throughout the sample keeps winning out of the sample, just as a real anomaly would, so these diagnostics cannot tell the two apart.

Empirical contribution 1 (data). We characterize the defect as far as the data allow. The evidence is consistent with a persistent, open-specific downward bias, concentrated in 2013 but present in every three-year window, and not with stale prints alone; the process that generated it is not identified. We also study the properties of a reference-free binomial screen for it: its justification, assumptions, size, power and false-alarm rate, in the settings we examined.

Empirical contribution 2 (search space). We document how a strategy's significance changes once the search space is declared, keeping coding errors, specification choices, and search decisions separate.

Practical contribution. We propose a four-stage reporting protocol (data validation, strategy specification, multiplicity correction, economic interpretation). Its multiplicity stage benchmarks every result against a Monte Carlo noise ceiling, reported together with its sampling uncertainty.

The empirical demonstrations are evidence for the core contribution; we do not claim

that the screen is a general test for data corruption, or that the absence of a profitable strategy extends beyond the instruments, data, strategy families, frequencies, and periods examined.

The remainder of the paper is organized as follows. Section 2 reviews the literature on data quality, multiple testing and backtest overfitting, intraday and overnight return patterns, and reproducibility. Section 3 describes the data and the statistical conventions used throughout. Section 4 reproduces the naive decomposition on the free-provider series. Section 5 defines the 24-strategy grid and Section 6 reports its performance. Section 7 analyzes transaction costs and Section 8 applies formal tests and the corrections for the search. Section 9 shows what the standard diagnostics say about the naive result, isolates the source of the effect in the opening prices, and studies a screen that uses only the series' own prices. Section 10 documents the second failure, an undeclared search space. Section 11 runs a fully declared search on the futures data and introduces the noise ceiling. Section 12 asks whether anything in this market is profitable when the analysis is driven by structure rather than by enumeration. Section 13 states the reporting protocol, Section 14 discusses the episode, Section 15 states the limitations, and Section 16 concludes. Appendices A to D document the search spaces, the resampling procedures, the computational environment, and the spread estimates.

2. Literature Review

This paper sits where two research streams meet: one on the quality of financial data, the other on inference after a specification search. We review each, then the literature on intraday and overnight return patterns that supplies the empirical setting, and finally, the work on reproducibility and reporting standards that our protocol builds on.

2.1. Data Quality in Financial Research

Errors in commercial and free financial databases have been documented for decades, and their effect on inference is often large relative to the effects being studied. Ince and Porter [8] show that unscreened Datastream equity returns contain enough errors to change conclusions about anomalies. Shumway [12] shows that missing delisting returns in CRSP bias measured small-firm returns upward, and Canina et al. [13] show that compounding the daily CRSP equal-weighted index produces a spurious drift through bid–ask bounce. Elton et al. [14] document discrepancies between two mutual-fund databases, and Ljungqvist et al. [15] show that historical analyst recommendations were changed retrospectively. In high-frequency data, cleaning procedures for outliers, mistimed records, and misreported quotes are a standard part of the pipeline [16–18]. Two features of this literature matter here. First, the documented defects usually concern returns, prices, or coverage in general, not a single field of an OHLC bar. Second, the remedy it proposes is data cleaning before estimation. That remedy is not linked to the overfitting diagnostics applied after estimation, and this paper shows what goes wrong when the two are treated separately.

2.2. Multiple Testing and Backtest Overfitting

Data snooping was formalised in finance by Lo and MacKinlay [9] and turned into resampling tests by White [1], Hansen [2] and Romano and Wolf [3]. Sullivan et al. [10] applied the Reality Check to technical trading rules, and later studies applied stepwise and false-discovery procedures to rules in equity and currency markets [19–21]. In the cross-section of expected returns, Harvey et al. [11] and Harvey and Liu [22] argue for higher significance thresholds, Chordia et al. [23] estimate the share of false rejections, Hou et al. [24] and Jensen et al. [25] reach different conclusions about replicability, and McLean and Pontiff [26] document post-publication decay. A parallel literature measures overfitting in

a backtest directly: the probabilistic and deflated Sharpe ratios [4,27], combinatorially symmetric cross-validation and the probability of backtest overfitting [5], a Bayesian alternative [28], and estimates of the effective number of independent trials by clustering [29]. Every one of these procedures conditions on the data and on the declared set of trials. Bailey et al. [30] and Fabozzi and López de Prado [31] stress that the number of trials must be reported honestly, but they do not address what happens when the data themselves are wrong. Section 9 shows that in that case, the diagnostics do not just lose power: they reverse their ranking of the comparator and defective series.

2.3. Overnight and Intraday Returns, and Currency Market Microstructure

The empirical setting is the overnight/daytime decomposition. In U.S. equities, most of the cumulative return has accrued overnight [32–34], and Salotra et al. [7] apply the framework used here to sector exchange-traded funds. In currencies, trading is nearly continuous, and daily bar boundaries are a convention of the data provider rather than of the market. Ranaldo [35] and Breedon and Ranaldo [36] document time-of-day patterns in which currencies tend to depreciate during their own domestic trading hours, and Andersen and Bollerslev [37] document strong intraday seasonality in currency volatility. These patterns make an overnight effect in the dollar economically plausible, which is why the spurious result in Section 4 is easy to rationalize. Microstructure also determines what an "open" is: tick size and price discreteness [38], bid–ask bounce [39] and the construction of the high and low [40–42] all affect where an observed open sits within the session range. Section 9.5 uses these results to state the null hypothesis of the proposed screen.

2.4. Reproducibility and Reporting Protocols

Concerns about reproducibility in empirical finance now extend beyond multiple testing. Menkveld et al. [43] show that independent teams testing the same hypotheses on the same data reach materially different results, and Harvey [44] argues for pre-specification and reporting of all tests. Arnott et al. [45] propose a backtesting protocol for machine-learning strategies and Fabozzi and López de Prado [31] propose a template for disclosing the number of tests. These protocols concern the research process and the declaration of trials. None of them includes a screening step for the input data, or a benchmark expressed as the performance the declared search attains on resampled data.

2.5. Positioning

The literature therefore offers data-cleaning procedures applied before estimation and multiplicity corrections applied after it, but little evidence on how they interact. Our core contribution (Section 9) shows on real data that a defect present throughout the sample receives a better overfitting score than the comparator series. Section 10 shows on the same instrument how an undeclared search space changes the denominator. Section 13 combines the two lessons into a protocol that adds a data-validation stage and a noise-ceiling benchmark to the existing reporting templates [31,45].

3. Data and Methods

3.1. The Index Is Not Directly Tradable

Before describing the four data sources used in this paper, we state a distinction that turns out to matter for everything that follows. The DXY index is a calculated benchmark: ICE Data Indices, its administrator, computes it from a real-time feed of the six constituent currencies' mid-point spot rates, and states directly that it is not possible to invest in the index itself [46]. There is no security whose price is the index level. Exposure is instead taken through the ICE U.S. Dollar Index futures contract, or through futures-based

exchange-traded products such as the Invesco DB US Dollar Index Bullish Fund, which itself holds long positions in the futures contract rather than in the index [47].

This distinction is not merely terminological, and it is the reason this paper exists. A free, widely used redistribution of the DXY *spot* index is easy to obtain and easy to mistake for a tradable series – indeed, an earlier pass of this analysis did exactly that, treating a popular free provider's DXY feed as if it were the futures contract. It is not: as Table 1 shows, the free series and the commercial exchange-sourced futures series are two different products, both nominally tracking the same index, obtained from two different pipelines. They are related but distinct instruments. They can differ in trading hours, in how the daily bar is defined, in price formation (an index calculated from spot mid-rates versus traded futures prices), in settlement, and in microstructure. The futures price additionally embeds the cost of carry between valuation and settlement [48], so even a correctly labeled futures series would not equal the spot index level exactly. We therefore do not treat the futures series as the "true" version of the spot index. We use it as an independently sourced comparator for the same sessions, and Section 9 examines the gap between the two in detail. The evidence shows that the gap is not carry, and it is consistent with inaccurate opening observations in the free series.

3.2. Data Sources

Table 1 lists the four sources. The two U.S. Dollar Index series are the center of the paper. One is the ICE spot index as redistributed by a free provider, the series on which the original result was found. The other is the ICE Dollar Index futures contract obtained from a commercial vendor sourced from the exchange. They describe the same underlying index over the same dates.

Table 1. Data sources. The two dollar index series in the first two rows describe the same underlying index over the same dates, and are the comparison on which Section 9 rests. The free index and the same provider's constituent spot bars are distributed with the replication code; file layouts, and processing are described in Appendix C.

Series	Source	Period	Sessions
Dollar index, free provider	Free redistribution of the ICE spot index	2010-12 to 2025-12	3,932
Dollar index futures	Commercial vendor, exchange sourced (DX)	2010-12 to 2025-12	3,889
Currency futures	CME contracts E6, B6, D6, J6, S6	2010-12 to 2025-12	3,800
Spot rates and interest rates	Federal Reserve H.10 and OECD via FRED	1999-01 to 2026-08	6,929

The five currency futures are the CME contracts for the euro, sterling, Canadian dollar, yen, and Swiss franc. No liquid Swedish krona contract exists, so the tradeable universe is five of the six index constituents. All five carry median daily volumes above 25,000 contracts, contain no violation of the open, high, low, close ordering and are correctly dated: their return correlation with the Federal Reserve noon fix peaks at lag zero.

Nearby continuous futures jump when the front contract changes. That jump is the spread between two contracts and not a return any holder earns, so all returns are computed within the contract, and the 61 roll sessions in each series are assigned a return of zero. This also means that contract changes cannot contribute to the overnight leg examined in Section 9.

3.3. Statistical Conventions

Every t statistic reported below is corrected for heteroskedasticity and autocorrelation following Newey and West [49], using a Bartlett kernel with the standard automatic bandwidth $\lfloor 4(T/100)^{2/9} \rfloor$. The bandwidth depends on the sample length T: it gives nine lags on the 3,932-session free-provider sample used in Sections 4–8 and eight lags on the 3,793 to 3,889 sessions of the comparisons in Section 9 and of the futures panel. This is why Table 12 uses nine lags and Section 9 uses eight. Table 12 also reports eight and twelve lags. The conclusions are insensitive to that choice: the overnight leg of the free series returns $p < 0.0001$ at eight, nine and twelve lags while the exchange series returns 0.328, 0.328, and 0.331.

We report two kinds of quantity and keep them apart. Conventional p-values (Newey–West tests, White's Reality Check, Hansen's SPA test, Monte Carlo p-values) are probabilities under a null hypothesis, and small values are evidence against it. The probabilistic Sharpe ratio (PSR) and the deflated Sharpe ratio (DSR) of Bailey and López de Prado [4,27] are *not* p-values. They estimate the probability that the true Sharpe ratio exceeds a benchmark: zero or buy-and-hold for the PSR, and the expected maximum Sharpe ratio of N unskilled trials for the DSR. Large values are evidence of skill, and following Bailey and López de Prado [4] a DSR must exceed 0.95 to be read as significant at the 5% level. A DSR of 0.046, therefore, means that skill is very unlikely, not that the result is significant.

We also distinguish two kinds of inference. Single-strategy tests (Newey–West, PSR, bootstrap confidence intervals) are valid only for a strategy chosen before seeing the data. Search-adjusted tests (DSR, Reality Check, SPA, Romano–Wolf, the probability of backtest overfitting, and the noise ceiling) account for the selection of the best of a declared set. Throughout the paper, results of the first kind are labeled as conditional on the strategy choice. The implementation settings of every resampling-based procedure (benchmark, performance differential, resampling scheme, block length, number of replications, and random seed) are listed in Table A3 in Appendix B.

3.4. Sub-Period Return Definitions

For each day d_i we distinguish two sub-periods. The overnight sub-period runs from the close of day d_{i-1} to the open of day d_i. The daytime sub-period runs from the open of day d_i to the close of day d_i. Following Salotra et al. [7], we denote the corresponding returns as

$$R_i^{CO} = \frac{O_i}{C_{i-1}} - 1, \tag{1}$$

$$R_i^{OC} = \frac{C_i}{O_i} - 1, \tag{2}$$

where O_i and C_i are the open and close on day d_i. The close-to-close return satisfies

$$1 + R_i^{CC} = (1 + R_i^{CO})(1 + R_i^{OC}), \tag{3}$$

so the decomposition is exhaustive and lossless. For a currency index that trades almost around the clock, "overnight" and "daytime" are defined by the data provider's bar boundaries rather than by a market closure. The free provider does not document its session definition, so where the two sources place the open relative to each other is estimated from the data in Section 9.3.

A position of USD 100 is compounded through the sub-period return series. The final balance B_i on day d_i follows from the strategy return R_i and the previous balance as $B_i = (1 + R_i)B_{i-1}$. We report the terminal balance, total return, compound annual growth

rate (CAGR), the annualized Sharpe ratio at a zero risk-free rate,

$$\mathrm{SR} = \frac{\bar{R}_{\mathrm{daily}}}{\sigma_{\mathrm{daily}}}\sqrt{252}, \tag{4}$$

annualized volatility, maximum drawdown, and the trade count. A trade is counted whenever the target position at the start of a sub-period differs from the position carried into it and is non-zero.

Transaction costs are charged as a proportional deduction of c basis points from the sub-period return on every trade, for $c \in \{0, 1, 2\}$. We stress that $c = 0$ is a diagnostic rather than a scenario: it isolates the statistical signal from the execution problem. For a liquid futures contract, as in the exchange-sourced analysis of Sections 9 onward, $c = 1$ and $c = 2$ are meant to bracket the costs commonly assumed for an institutional participant and for a smaller one. These levels are sensitivity assumptions, not measured institutional costs. They do not model the bid–ask spread at the specific execution times, slippage relative to the recorded open or close, market impact, or whether the recorded prices were available to trade at all. Appendix D reports OHLC-based spread estimates for the futures and discusses why they bound rather than measure these costs. Section 4 applies the same convention to the free-provider series as a diagnostic benchmark, without implying that the series is itself tradable.

3.5. Software, Reproducibility and Generative AI Disclosure

All analysis was performed in Python 3 using `pandas`, `NumPy`, `SciPy` and `statsmodels`. Figures were produced with `matplotlib`. Appendix C documents the computational environment, package versions, random seeds, run times, the mapping from scripts to tables and figures, and the provenance of the code repository.

During the preparation of this manuscript, the authors used a large language model (Claude 4.6) to verify the statistical computations. The authors independently verified all analytical output, re-ran every reported figure from the raw data, and take full responsibility for the content of this publication.

4. The Naive Decomposition

A growing body of evidence shows that the overnight and daytime portions of a trading day carry different risk premia. In U.S. equities, the overnight session has historically earned the bulk of the market's cumulative return, whereas the daytime session contributed comparatively little [32–34]. Whether an analogous, or opposite, pattern exists in currency markets motivated the exercise reproduced in this section: a complete sub-period decomposition of the DXY on the free-provider series described in Table 1, run exactly as an analyst without a second, exchange-sourced series to cross-check against would run it. Every number in this section is computed on that series. We present the full apparatus – worked example, complete strategy grid, transaction cost decay, and formal statistical testing – deliberately, because the point of the next section is not that this analysis is careless. It is methodologically ordinary, applies established diagnostics correctly, and still produces a result that does not appear on the independently sourced comparator.

4.1. A Worked Example

Before turning to the full sample, it is useful to see the mechanics on real data. Table 2 gives the DXY open and close for the week of 27 June 2016, and Table 3 converts those prices into the three returns defined above. The week is a compact illustration of the paper's central fact: compounded across the five sessions, the overnight leg loses 0.632%, the daytime leg gains 0.847%, and the 24 h return a passive holder actually receives is the

small residual of the two, +0.210%.

Table 2. Example of free-provider DXY data. Daily open O_i and close C_i for the week of 27 June 2016. The close of the preceding session, 24 June 2016, is included because the first overnight return is measured from it.

Date d_i	**Day**	**Open** O_i	**Close** C_i
24 June 2016	Friday	93.45	95.45
27 June 2016	Monday	95.53	96.54
28 June 2016	Tuesday	96.41	96.25
29 June 2016	Wednesday	96.04	95.77
30 June 2016	Thursday	95.67	96.14
1 July 2016	Friday	95.89	95.65

Table 3. The same week decomposed into sub-period returns (%). R_i^{CO} is the overnight (close-to-open) return, R_i^{OC} the daytime (open-to-close) return, and R_i^{CC} the 24 h return, which satisfies $1 + R_i^{CC} = (1 + R_i^{CO})(1 + R_i^{OC})$. Over this week the overnight leg loses 0.63% in total while the daytime leg gains 0.85%, the asymmetry this paper is about.

		Prices		**Returns (%)**		
Date d_i	**Day**	**Open** O_i	**Close** C_i	R_i^{CO} **Night**	R_i^{OC} **Day**	R_i^{CC} **24 h**
27 Jun	Mon	95.53	96.54	0.084	1.057	1.142
28 Jun	Tue	96.41	96.25	−0.135	−0.166	−0.300
29 Jun	Wed	96.04	95.77	−0.218	−0.281	−0.499
30 Jun	Thu	95.67	96.14	−0.104	0.491	0.386
1 Jul	Fri	95.89	95.65	−0.260	−0.250	−0.510
Compounded over the week				−0.632	0.847	0.210

Table 4. Positions held by five key strategies in each sub-period of the example week. The figures beneath each day give that day's realized R_i^{CO} and R_i^{OC} in %. Inertia and reversal rules condition only on information already observable: the overnight position uses the previous daytime return, the daytime position uses the same morning's overnight return.

		Mon		**Tue**		**Wed**		**Thu**		**Fri**	
#	**Strategy**	**Night** 0.08	**Day** 1.06	**Night** −0.13	**Day** −0.17	**Night** −0.22	**Day** −0.28	**Night** −0.10	**Day** 0.49	**Night** −0.26	**Day** −0.25
5	(Long, Long)	long	long	long	long	long	long	long	long	long	long
2	(Short, Cash)	short	cash	short	cash	short	cash	short	cash	short	cash
3	(Cash, Long)	cash	long	cash	long	cash	long	cash	long	cash	long
7	(Short, Long)	short	long	short	long	short	long	short	long	short	long
20	(Short, Reversal)	short	short	short	long	short	long	short	long	short	long

Table 5. Accumulation of USD 100 through the example week, at zero transaction cost. Each day's balance is $B_i = (1 + R_i)B_{i-1}$ with $R_i = (1 + p_i^{CO} R_i^{CO})(1 + p_i^{OC} R_i^{OC}) - 1$, where p is the position from Table 4. Strategy 5 is buy-and-hold in sub-period form.

#	**Strategy**	**Start**	**Mon**	**Tue**	**Wed**	**Thu**	**Fri**
5	(Long, Long)	100.00	101.14	100.84	100.34	100.72	100.21
2	(Short, Cash)	100.00	99.92	100.05	100.27	100.37	100.63
3	(Cash, Long)	100.00	101.06	100.89	100.61	101.10	100.85
7	(Short, Long)	100.00	100.97	100.94	100.88	101.48	101.49
20	(Short, Reversal)	100.00	98.86	98.83	98.77	99.35	99.36

Table 4 shows the position each of five representative strategies holds in each sub-period, and Table 5 compounds USD 100 forward through them. The ordering over this

single week already previews the full sample. Strategy 7, which shorts the overnight leg and holds the daytime leg, ends at USD 101.49 against USD 100.21 for buy-and-hold, and it does so while being flat on the direction that hurts the passive holder. Strategy 20 ends below 100 because its daytime reversal rule was wrong on the Monday, when a positive overnight return led it to take a short position into a session that rose 1.06%. One week is, of course, not evidence of anything; it is shown so that every number in the tables that follow can be traced back to a transparent construction.

4.2. Data Limitation

The free-provider DXY series contains 138 days, or 3.5% of the sample, on which the series did not print while the spot currency pairs did. We treat those days as flat across all strategies. The buy-and-hold benchmark is computed using the forward-filled close series to reflect the true price ratio $98.24/80.71 = 1.2172$. This works against the strategies that forgo moves for the missing days. Three terminal values therefore appear in the paper, and they differ only in how the 138 gaps are handled. (i) Buy-and-hold on the forward-filled close, USD 121.72, earns every price change, including the move across each gap. (ii) Strategy 5, (Long, Long), in Table 6, USD 119.61, is the sub-period strategy under the flat-on-missing-days convention: on the session after a gap the overnight return $O_i/C_{i-1} - 1$ is undefined because C_{i-1} is missing, so it is set to zero, while that session's daytime return $C_i/O_i - 1$ is still earned. The difference from (i) is the 138 forgone overnight moves across the gaps. (iii) A close-to-close benchmark under the same skipped-day convention, USD 104.76, loses more because a missing close makes two close-to-close returns undefined (the gap day and the day after), so both the overnight and the daytime part of the post-gap session are forgone, 276 returns in total. We use the highest and hardest-to-beat figure, USD 121.72, for buy-and-hold throughout, and Strategy 5's USD 119.61 as the passive reference within the sub-period grid, because it is subject to exactly the same convention as the other 23 strategies.

5. Strategies

Each of the two daily sub-periods is assigned one of five elementary rules:

- **Cash**: hold no position;
- **Long**: hold a long position unconditionally;
- **Short**: hold a short position unconditionally;
- **Inertia**: take a long (short) position if the return for the preceding sub-period was non-negative (negative);
- **Reversal**: take a long (short) position if the return for the preceding sub-period was negative (non-negative).

We denote a daily strategy as a pair (overnight, daytime). There are $5 \times 5 = 25$ possible combinations. Excluding the trivial (Cash, Cash) strategy, there are 24 strategies to consider, so that we are always invested in at least one sub-period. All conditioning information is strictly backward-looking: the overnight position for day d_i is set from the daytime return on day d_{i-1}, and the daytime position for day d_i is set from the overnight return on day d_i, which is already observable at the open. The 24 strategies are enumerated in Table 6.

The family is the one used by Salotra et al. [7] for sector exchange-traded funds, so it is fixed before the data are examined rather than discovered by looking at the data. The rationale for the five rules is as follows. The two unconditional rules test whether either sub-period carries a premium. Inertia tests whether a move in one sub-period continues into the next, as a delayed-adjustment or order-flow account would predict. Reversal tests whether it is undone, as a liquidity-provision or overreaction account would predict. Cash isolates each leg. Pairing the rules across the two legs gives every combination of these

hypotheses, so the grid is a complete design over one small set of ideas rather than a search over unrelated signals.

6. Strategy Performance Analysis

6.1. The Dollar Earns Its Return During the Daytime Sub-Period

Figure 1 presents the central empirical fact on the free-provider series. Holding DXY only during the open-to-close sub-period grows USD 100 into USD 196.01, a CAGR of 4.41%, a Sharpe ratio of 0.68 and a maximum drawdown of −12.4%. Holding it only overnight shrinks USD 100 to USD 61.02, a CAGR of −3.12% and a Sharpe ratio of −1.43. Buy-and-hold, which is the product of the two legs, ends at USD 121.72.

The overnight leg is notable less for its magnitude than for its consistency. Its annualized volatility is only 2.19%, roughly one-third that of the daytime leg, yet its Sharpe ratio of −1.43 is the largest in absolute value in our set. The mirror strategy, (Short, Cash), delivers a Sharpe ratio of +1.43 with a maximum drawdown of only −5.3%.

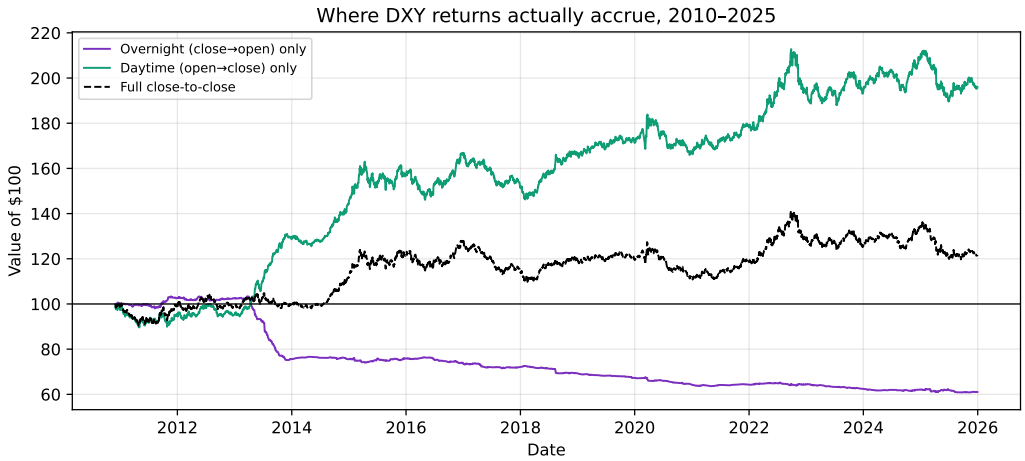


Figure 1. Cumulative value of USD 100 invested in the DXY overnight sub-period only, the daytime sub-period only, and continuously, 2010–2025, on the free-provider series. The two sub-periods move in opposite directions across the entire sample.

6.2. The Full Strategy Set

Table 6 reports all 24 strategies at zero transaction cost. The best performer is Strategy 7, (Short, Long), which combines the two legs identified above. It ends at USD 318.80, a total return of 218.8%, a CAGR of 7.71%, a Sharpe ratio of 1.06, and a maximum drawdown of −13.0%. Figure 2 plots the equity curves of the leading strategies and Figure 5 the drawdown profile of Strategy 7 against buy-and-hold.

6.3. Mirror-Pair Structure and Effective Number of Trials

The strategy set is not 24 independent experiments. Because each elementary rule has an exact negation within the set, the strategies partition into 12 mirror pairs whose Sharpe ratios are equal or near equal in magnitude and opposite in sign. Pair (1, 2) sits at exactly ∓1.432 and pair (3, 4) at exactly ±0.681. Pair (7, 8) comes out at +1.064 and −1.061, and pair (17, 20) at −0.850 and +0.853. Pairs involving a Cash leg mirror exactly, whereas pairs with two active legs differ in the third decimal because compounding two sub-period returns is not a linear operation.

The first submission treated the grid as roughly 12 sign choices. As the reviewers point out, mirror pairs alone do not justify halving the number of trials, because the dependence between strategies extends beyond the pairs. We therefore estimate the effective number of

Table 6. All 24 strategies on DXY, 2010–2025, at zero transaction cost, free-provider series. "Final" is the terminal value of USD 100. Strategies are ordered by construction, not by performance. The buy-and-hold row uses the forward-filled close and Strategy 5 the flat-on-missing-days convention common to all 24 strategies (Section 4.2).

#	Overnight	Daytime	Final	Total %	CAGR %	Sharpe	MaxDD %	Vol %	Trades
1	Long	Cash	61.02	−39.0	−3.12	−1.432	−41.2	2.19	3932
2	Short	Cash	162.64	62.6	3.17	1.432	−5.3	2.19	3932
3	Cash	Long	196.01	96.0	4.41	0.681	−12.4	6.66	3932
4	Cash	Short	47.60	−52.4	−4.65	−0.681	−60.3	6.66	3932
5	Long	Long	119.61	19.6	1.15	0.203	−14.3	6.80	1
6	Short	Short	77.41	−22.6	−1.63	−0.207	−38.5	6.80	1
7	**Short**	**Long**	**318.80**	**218.8**	**7.71**	**1.064**	**−13.0**	**7.23**	**7864**
8	Long	Short	29.04	−71.0	−7.62	−1.061	−75.8	7.22	7864
9	Cash	Inertia	61.20	−38.8	−3.10	−0.448	−54.9	6.54	3656
10	Cash	Reversal	152.83	52.8	2.76	0.448	−32.0	6.54	3656
11	Inertia	Cash	77.98	−22.0	−1.58	−0.713	−28.4	2.20	3793
12	Reversal	Cash	127.26	27.3	1.56	0.713	−7.9	2.20	3793
13	Inertia	Inertia	47.73	−52.3	−4.63	−0.643	−63.2	6.99	3881
14	Inertia	Reversal	119.18	19.2	1.13	0.199	−38.9	6.81	3706
15	Reversal	Inertia	77.89	−22.1	−1.59	−0.201	−47.2	6.81	3706
16	Reversal	Reversal	194.50	94.5	4.36	0.644	−28.0	7.00	3881
17	Long	Inertia	37.35	−62.7	−6.12	−0.850	−71.3	7.13	3894
18	Long	Reversal	93.26	−6.7	−0.45	−0.034	−44.7	6.66	3971
19	Short	Inertia	99.54	−0.5	−0.03	0.029	−38.6	6.66	3971
20	Short	Reversal	248.56	148.6	6.01	0.853	−29.4	7.14	3894
21	Inertia	Long	152.86	52.9	2.76	0.432	−15.1	6.83	3737
22	Reversal	Long	249.46	149.5	6.03	0.849	−13.7	7.20	4127
23	Inertia	Short	37.12	−62.9	−6.15	−0.846	−69.9	7.20	4127
24	Reversal	Short	60.57	−39.4	−3.16	−0.436	−51.1	6.83	3737
DXY buy-and-hold			121.72	21.7	1.27	0.218	−15.3	6.86	1

independent trials formally from the 24×24 correlation matrix of daily strategy returns. Every strategy's daily return is, to first order, a signed combination of four streams: the overnight return, the daytime return, the overnight return signed by the previous daytime return, and the daytime return signed by the same day's overnight return. The correlation matrix is therefore close to rank four. Its four largest eigenvalues account for 40.0%, 37.1%, 14.8% and 8.0% of the total, and the remaining twenty for less than 0.001%. Table 7 reports four standard estimators of the effective number of tests built on this spectrum or on clustering, together with the deflated Sharpe ratio each implies.

Table 7. Effective number of independent trials in the 24-strategy grid (free-provider series) and the deflated Sharpe ratio implied by each estimate. Nyholt, Li–Ji and Galwey are eigenvalue-based estimators [50–52]. The clustering estimate cuts an average-linkage dendrogram on $1 - |\rho|$ at $|\rho| = 0.5$, so that a mirror pair counts as one trial, in the spirit of López de Prado and Lewis [29]. $E[\max]$ is the expected maximum annualised Sharpe ratio of N unskilled trials with the observed cross-sectional dispersion. The DSR is a probability and must exceed 0.95 (Section 3.3). Input to every estimator: the 24×24 Pearson correlation matrix of daily strategy returns at zero cost on the 3,932 free-provider sessions. Every DSR uses the same $E[\max]$ formula and the same variance of the 24 observed Sharpe ratios; only N differs between rows. No row is the true number of independent trials.

Estimate of N	N	$E[\max]$	**DSR, S7**	**DSR, S2**	**DSR, S22**
Nominal	24.0	1.486	0.046	0.405	0.006
Nyholt [50]	17.2	1.376	0.107	0.597	0.018
Mirror pairs (first submission)	12.0	1.250	0.230	0.789	0.056
Li and Ji [51]	7.0	1.041	0.537	0.958	0.223
Clusters at $\|\rho\| > 0.5$	4.0	0.790	0.863	0.998	0.594
Galwey [52]	3.7	0.745	0.898	0.999	0.660

Assumptions of the estimators. All four use the same input and differ only in how they

summarise the eigenvalues $\lambda_1, \ldots, \lambda_M$ ($M = 24$) of the correlation matrix, or its cluster structure. Nyholt's estimator is $1 + (M-1)\big(1 - \mathrm{Var}(\lambda)/M\big)$, Li and Ji's is $\sum_i \big[\mathbb{1}(\lambda_i \geq 1) + (\lambda_i - \lfloor \lambda_i \rfloor)\big]$, and Galwey's is $(\sum_i \sqrt{\lambda_i})^2 / \sum_i \lambda_i$. They were developed for correlated tests in genetic association studies, where the correlation matrix of the tests is taken as a summary of their dependence. None was designed for the maximum of a set of Sharpe ratios. The clustering estimate depends on its cut-off: at $|\rho| = 0.3$ it gives three clusters and at $|\rho| = 0.7$ four. The deflated Sharpe ratio then assumes N independent trials whose Sharpe ratios have the variance observed across all 24, so a smaller N lowers $E[\max]$ but leaves that variance unchanged.

The estimates range from 3.7 to 17.2, so the effective number of trials is far from settled, and we do not choose among them. The estimators disagree because the spectrum is dominated by four streams of very unequal size. Two conclusions are nevertheless robust. Under every estimate, Strategy 7, the headline combination, has a DSR below 0.95. Strategy 2 (the overnight leg alone) has a DSR of 0.41 to 0.79 under the three largest values of N (24, 17.2 and 12) and 0.96 to 1.00 under the three smallest (7.0, 4 and 3.7), where it would be read as a significant discovery. Whether the DSR supports the overnight leg therefore depends on which estimate of N is adopted, and Section 9 shows that this leg is associated with the free series' recorded opening prices rather than with a market effect. The resampling-based tests in Section 8 account for the dependence directly through the joint bootstrap, so they do not need an estimate of N.

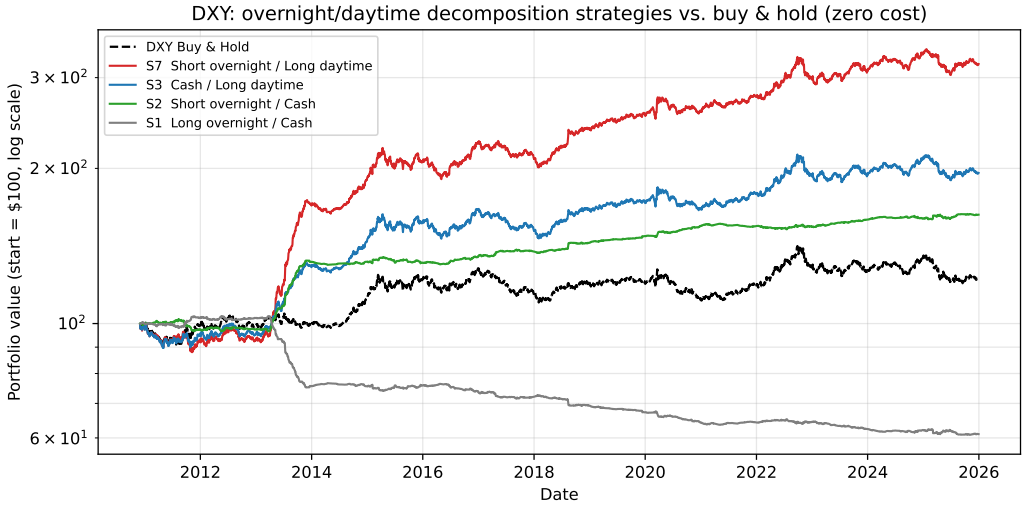


Figure 2. Equity curves for the leading strategies against buy-and-hold, log scale, at zero transaction cost, free-provider series.

6.4. Extension to the Constituent Currency Pairs

Applying the same 24-strategy set to each of the six DXY constituents provides a useful cross-check. The best single pair and strategy combination at zero cost is USD/JPY under Strategy 8, (Long, Short), at USD 170.96 with a Sharpe ratio of 0.42. In the first submission we read this, together with the fact that no constituent comes within 150 percentage points of the index-level Strategy 7, as evidence that the index effect was partly inherited from the yen leg and partly produced by diversification across the six pairs. That reading was not supported by a formal decomposition, and the decomposition reported in Section 9.3 (Figure 6) contradicts it. Under the index's own log-linear formula, the free constituents' overnight legs sum to a cumulative log return of $+0.096$, against -0.482 for the free index, and the yen leg contributes $+0.075$, the opposite sign to the index effect. Neither inheritance from a constituent nor diversification explains the index result, and we withdraw that

claim.

7. Transaction Cost Analysis

Strategy 7 changes position twice per day, giving 7864 trades over 3932 sessions. Table 8 and Figure 3 show the consequence.

Table 8. Terminal value of USD 100 for all 24 strategies at three per-trade cost levels, free-provider series. Blue marks Strategy 5, (Long, Long), which is buy-and-hold expressed in sub-period form and is the passive reference. Green marks a strategy that ends above the reference, red one that ends below it, and a yellow one that ends within 10% below the reference but on a higher Sharpe ratio. The yellow category is used only in this table: it flags strategies that lose to the reference in money terms but beat it on a risk-adjusted basis, a distinction that does not arise for the single-metric comparisons in Tables 9–11, which use green and red only.

#	Overnight	Daytime	0 bps	1 bps	2 bps
1	Long	Cash	61.02	42.33	29.37
2	Short	Cash	162.64	112.84	78.29
3	Cash	Long	196.01	134.13	91.78
4	Cash	Short	47.60	32.57	22.28
5	Long	Long	119.61	119.61	119.61
6	Short	Short	77.41	77.41	77.41
7	Short	Long	318.80	151.36	71.85
8	Long	Short	29.04	13.79	6.54
9	Cash	Inertia	61.20	42.46	29.45
10	Cash	Reversal	152.83	106.03	73.56
11	Inertia	Cash	77.98	54.10	37.53
12	Reversal	Cash	127.26	88.29	61.25
13	Inertia	Inertia	47.73	32.59	22.25
14	Inertia	Reversal	119.18	82.86	57.60
15	Reversal	Inertia	77.89	54.15	37.64
16	Reversal	Reversal	194.50	132.81	90.69
17	Long	Inertia	37.35	25.83	17.87
18	Long	Reversal	93.26	64.00	43.91
19	Short	Inertia	99.54	68.31	46.88
20	Short	Reversal	248.56	171.96	118.96
21	Inertia	Long	152.86	106.48	74.17
22	Reversal	Long	249.46	167.69	112.73
23	Inertia	Short	37.12	24.95	16.77
24	Reversal	Short	60.57	42.19	29.38

Table 8 colors every cell against Strategy 5, the passive (Long, Long) reference, following the convention of Salotra et al. [7]. Nine strategies beat it at zero cost, five at 1 bps, and none at 2 bps. The single yellow cell is Strategy 2 at 1 bps: it ends at USD 112.84 against the reference's USD 119.61, so it loses money, yet its Sharpe ratio of 0.364 is almost double the reference's 0.203. That cell is the entire argument of Section 7 in one place.

1. **The best strategy changes with the cost assumption.** At 0 bps the best strategy is Strategy 7 at USD 318.80. At 1 bps it is Strategy 20, (Short, Reversal), at USD 171.96, because it trades half as often. At 2 bps, the best strategy in the entire set is Strategy 5, which is buy-and-hold expressed in sub-period form and trades once. Any claim about the best strategy is therefore conditional on an assumption about execution and should be stated as such.
2. **Strategy 7's edge is approximately 1.56 bps per trade.** Its terminal value falls from USD 318.80 to USD 151.36 to USD 71.85 as cost rises from 0 to 1 to 2 basis points. It drops below the buy-and-hold benchmark of USD 121.72 at approximately 1.29 bps and below break-even at approximately 1.56 bps, as shown in Figure 4. At the time, treating this as a liquid tradable futures contract, 1.56 bps looked like a narrow

but not obviously unachievable margin – which is why the accurate statement, even setting aside the data defect documented in Section 9, is that the naive analysis placed the strategy at the boundary of viability rather than reporting a clean 218.8% return. Section 9 shows the boundary question never actually arises, because the underlying series is not a tradable futures contract in the first place. Even for a tradable contract, a break-even of 1.56 bps per trade could not be assessed from the cost levels used here. It would require the quoted spread at the two execution times each day, slippage relative to the recorded open and close (which the evidence of Section 9 suggests may not be tradable prices), and market impact.

3. **Low volatility is not the same as robustness.** Strategy 2, (Short, Cash), has the highest gross Sharpe ratio in the entire set at 1.432, on the lowest volatility of 2.19% and the shallowest drawdown of −5.3%, and it trades once per day rather than twice. It is tempting to conclude that it is the safer implementation. It is not. Because a fixed per-trade cost is deducted in return space, it consumes a much larger fraction of a low-volatility strategy's return. At 1 bps, Strategy 2's Sharpe ratio falls to 0.364, below Strategy 7's 0.404, and at 2 bps it falls to −0.704 against Strategy 7's −0.257. The strategy with the best gross risk-adjusted profile is the one most damaged by frictions. Ranking strategies by gross Sharpe ratio, therefore, systematically favors precisely those strategies least able to pay for their own execution.

7.1. Risk Metrics Under Cost

Terminal value is only one of the things a cost deduction moves. Tables 9, 10 and 11 report the annualized Sharpe ratio, annualized volatility, and maximum drawdown at the same three cost levels. Each cell is colored against the same passive reference, Strategy 5, using a binary green/red comparison on the single metric shown. The yellow category of Table 8 is not needed here, because it exists only to flag a disagreement between the money and the risk-adjusted rankings. Reading them together makes the point of observation 3 above visible rather than merely asserted.

Two patterns stand out. First, volatility is almost inert to cost: ten strategies sit below the reference's 6.80% at zero cost, and eleven do so at 1 and 2 bps, and the column-to-column changes are in the third decimal. A fixed per-trade deduction shifts the mean of the return series without materially changing its dispersion. Second, Sharpe ratios and drawdowns are not inert at all. Nine strategies beat the reference's Sharpe ratio at zero cost, six at 1 bps, and none at 2 bps; the count of strategies with a shallower drawdown falls from five to one to zero over the same range. The strategies that lose their green first are, in every case, the ones that trade twice a day.

The clearest single case is Strategy 2, (Short, Cash). At zero cost, it has the highest Sharpe ratio in the set at 1.432, the lowest volatility at 2.19% and the shallowest drawdown at −5.3%. It is the safest-looking strategy in the table on every risk measure at once. By 2 bps its Sharpe ratio is −0.704, worse than Strategy 7's −0.257, and its drawdown has deepened from −5.3% to −31.8%. Its volatility, meanwhile, has not moved: 2.19% at every cost level. The risk did not increase. The return available to pay for the risk disappeared, and a low-volatility strategy has proportionally less of it to give away.

8. Robustness and Statistical Validation

A terminal balance is not evidence. This section asks whether the results above survive formal testing. In one important respect they do not.

Table 9. Annualized Sharpe ratio for all 24 strategies at three per-trade cost levels, free-provider series. Blue marks Strategy 5, (Long, Long), which is buy-and-hold expressed in sub-period form and is the passive reference. Green indicates a higher Sharpe ratio than the reference; red, a lower one.

#	Overnight	Daytime	0 bps	1 bps	2 bps
1	Long	Cash	−1.432	−2.498	−3.561
2	Short	Cash	1.432	0.364	−0.704
3	Cash	Long	0.681	0.316	−0.049
4	Cash	Short	−0.681	−1.045	−1.410
5	Long	Long	0.203	0.203	0.203
6	Short	Short	−0.207	−0.207	−0.207
7	Short	Long	1.064	0.404	−0.257
8	Long	Short	−1.061	−1.722	−2.384
9	Cash	Inertia	−0.448	−0.806	−1.164
10	Cash	Reversal	0.448	0.090	−0.268
11	Inertia	Cash	−0.713	−1.777	−2.839
12	Reversal	Cash	0.713	−0.352	−1.416
13	Inertia	Inertia	−0.643	−0.990	−1.335
14	Inertia	Reversal	0.199	−0.143	−0.487
15	Reversal	Inertia	−0.201	−0.542	−0.881
16	Reversal	Reversal	0.644	0.295	−0.055
17	Long	Inertia	−0.850	−1.178	−1.504
18	Long	Reversal	−0.034	−0.397	−0.761
19	Short	Inertia	0.029	−0.333	−0.692
20	Short	Reversal	0.853	0.523	0.192
21	Inertia	Long	0.432	0.093	−0.246
22	Reversal	Long	0.849	0.496	0.143
23	Inertia	Short	−0.846	−1.199	−1.550
24	Reversal	Short	−0.436	−0.775	−1.112

Table 10. Annualized volatility (%) for all 24 strategies at three per-trade cost levels, free-provider series. Blue marks Strategy 5, (Long, Long), which is buy-and-hold expressed in sub-period form and is the passive reference. Green marks lower volatility than the reference, red higher volatility.

#	Overnight	Daytime	0 bps	1 bps	2 bps
1	Long	Cash	2.19	2.20	2.20
2	Short	Cash	2.19	2.19	2.19
3	Cash	Long	6.66	6.66	6.66
4	Cash	Short	6.66	6.66	6.66
5	Long	Long	6.80	6.80	6.80
6	Short	Short	6.80	6.80	6.80
7	Short	Long	7.23	7.23	7.23
8	Long	Short	7.22	7.22	7.22
9	Cash	Inertia	6.54	6.54	6.54
10	Cash	Reversal	6.54	6.54	6.54
11	Inertia	Cash	2.20	2.20	2.20
12	Reversal	Cash	2.20	2.20	2.20
13	Inertia	Inertia	6.99	7.01	7.03
14	Inertia	Reversal	6.81	6.80	6.79
15	Reversal	Inertia	6.81	6.82	6.84
16	Reversal	Reversal	7.00	6.99	6.97
17	Long	Inertia	7.13	7.15	7.17
18	Long	Reversal	6.66	6.65	6.64
19	Short	Inertia	6.66	6.68	6.69
20	Short	Reversal	7.14	7.12	7.11
21	Inertia	Long	6.83	6.83	6.83
22	Reversal	Long	7.20	7.20	7.20
23	Inertia	Short	7.20	7.20	7.21
24	Reversal	Short	6.83	6.84	6.85

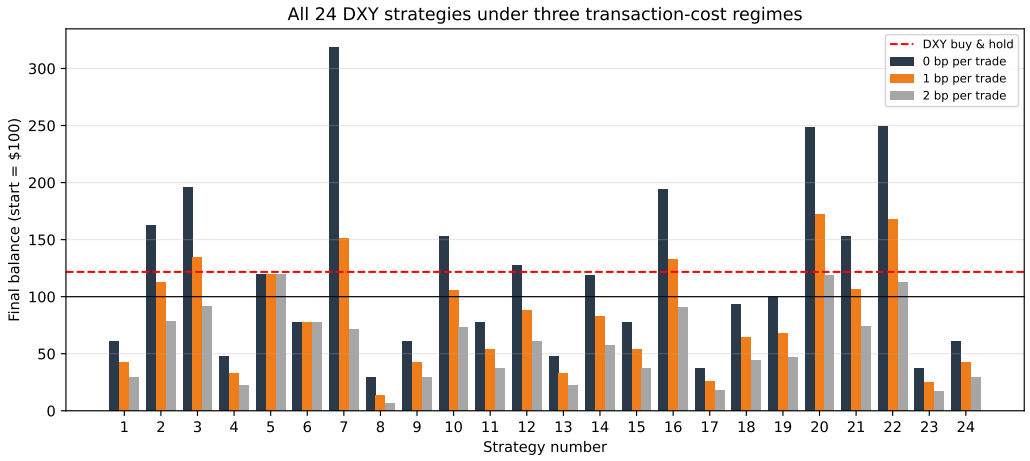


Figure 3. Terminal value of USD 100 for all 24 strategies at 0, 1, and 2 basis points per trade, free-provider series. The dashed line is DXY buy-and-hold.

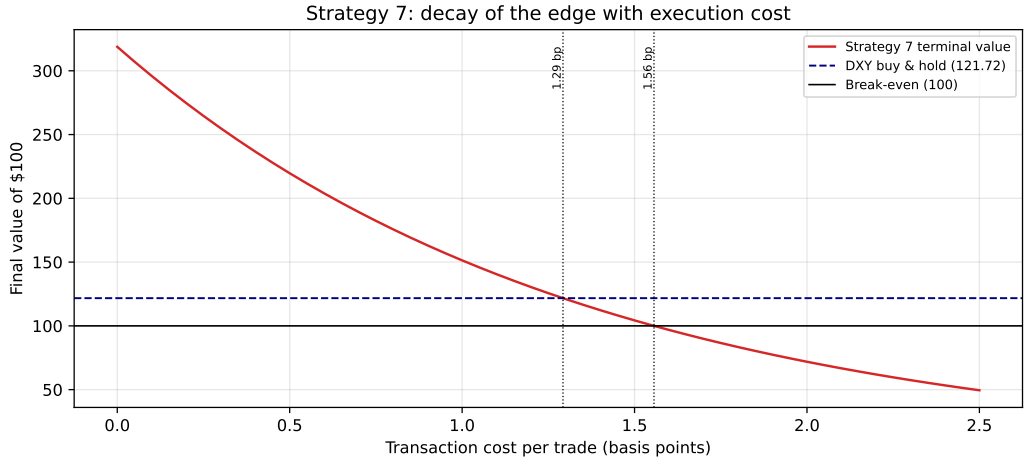


Figure 4. Terminal value of Strategy 7 as a continuous function of per-trade cost, free-provider series. The edge falls below buy-and-hold at 1.29 bps and below break-even at 1.56 bps.

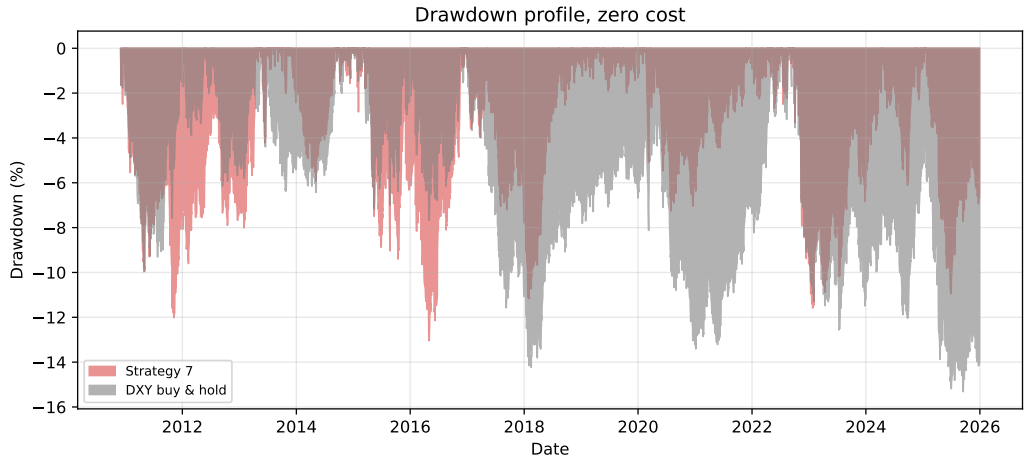


Figure 5. Drawdown profile of Strategy 7 against buy-and-hold at zero cost, free-provider series. Strategy 7's maximum drawdown of −13.0% is slightly shallower than the benchmark's −15.3% despite a 7.71% CAGR.

Table 11. Maximum drawdown (%) for all 24 strategies at three per-trade cost levels, free-provider series. Blue marks Strategy 5, (Long, Long), which is buy-and-hold expressed in sub-period form and is the passive reference. Green marks a shallower drawdown than the reference, red a deeper one.

#	Overnight	Daytime	0 bps	1 bps	2 bps
1	Long	Cash	−41.2	−58.0	−70.8
2	Short	Cash	−5.3	−9.1	−31.8
3	Cash	Long	−12.4	−16.9	−30.5
4	Cash	Short	−60.3	−71.9	−80.1
5	Long	Long	−14.3	−14.3	−14.3
6	Short	Short	−38.5	−38.5	−38.5
7	Short	Long	−13.0	−20.2	−49.8
8	Long	Short	−75.8	−87.5	−93.9
9	Cash	Inertia	−54.9	−64.7	−72.4
10	Cash	Reversal	−32.0	−46.5	−58.2
11	Inertia	Cash	−28.4	−48.1	−63.2
12	Reversal	Cash	−7.9	−16.7	−38.8
13	Inertia	Inertia	−63.2	−71.3	−77.9
14	Inertia	Reversal	−38.9	−52.5	−63.1
15	Reversal	Inertia	−47.2	−57.4	−66.5
16	Reversal	Reversal	−28.0	−40.5	−53.6
17	Long	Inertia	−71.3	−77.6	−82.6
18	Long	Reversal	−44.7	−56.9	−66.4
19	Short	Inertia	−38.6	−46.8	−57.7
20	Short	Reversal	−29.4	−36.4	−49.1
21	Inertia	Long	−15.1	−20.6	−36.7
22	Reversal	Long	−13.7	−16.1	−24.5
23	Inertia	Short	−69.9	−78.9	−85.2
24	Reversal	Short	−51.1	−63.6	−73.7

8.1. Are the 24 Strategies Distinguishable?

A one-way ANOVA on mean daily return across the 24 strategies rejects equality of means, with $F(23, 94344) = 6.97$ and $p < 0.0001$. However, every assumption behind that test fails on this data. Levene's test rejects equal variances at $p < 10^{-16}$, Jarque–Bera rejects normality of the winning strategy's returns with $JB = 2341.7$ at $p < 10^{-16}$, and Ljung–Box rejects independence at ten lags with $p < 0.0001$. We therefore also report the distribution-free Kruskal–Wallis analog, which agrees: $H = 113.15$ at $p < 0.0001$.

We include the ANOVA for completeness and to state explicitly that it is the wrong instrument for this question. It establishes only that the strategies differ from one another, which is guaranteed by construction, since twelve of them are the exact negations of the other twelve.

8.2. Is the Best Strategy's Mean Return Significantly Positive?

Table 12 reports Newey–West heteroskedasticity and autocorrelation consistent t-statistics on mean daily returns, using the automatic bandwidth, which is nine lags at this sample length (Section 3.3), and a Bartlett kernel [49], together with the Probabilistic Sharpe Ratio of Bailey and López de Prado [27] and confidence intervals from a stationary block bootstrap using 5000 resamples and a mean block length of 21 sessions (seed 1; Table A3). All quantities in Table 12 except the DSR condition on the strategy having been chosen in advance. They are reported for that reason and are not evidence of significance for a strategy selected as the best of the grid.

Read in isolation, these are strong results. Strategy 7's mean daily return is significant at $p = 0.0002$ after correcting for autocorrelation and heteroskedasticity, and its bootstrapped 95% confidence interval for the annualized Sharpe ratio is $[0.430, 1.765]$, comfortably excluding zero. The buy-and-hold benchmark, by contrast, is not statistically distinguishable from zero at all, with $t = 0.891$ and $p = 0.373$. Strategy 2's interval is

Table 12. Formal tests on the leading strategies, free-provider series. t_{NW} is the Newey–West t-statistic on mean daily return at nine lags (eight and twelve lags change no conclusion: t for Strategy 7 is 3.715 and 3.597). p is the two-sided p-value of the null hypothesis that the mean daily return is zero. PSR_b is the probability that the true Sharpe ratio exceeds the benchmark's 0.218. DSR is the Deflated Sharpe Ratio, correcting for 24 trials, skewness, kurtosis, and sample length; it is a probability and must exceed 0.95 (Section 3.3). The buy-and-hold row is the benchmark itself, so it has no strategy number and no PSR_b or DSR against itself. The first submission omitted PSR_b and DSR for Strategy 3 by mistake; they are now reported. SR, t_{NW}, p and PSR_b are conditional on the strategy having been chosen in advance; only the DSR adjusts for the search over 24 strategies.

#	Strategy	SR	t_{NW}	p	PSR_b	DSR
7	(Short, Long)	1.064	3.681	0.0002	0.9996	**0.046**
2	(Short, Cash)	1.432	4.317	1.6×10^{-5}	1.0000	0.405
22	(Reversal, Long)	0.849	3.104	0.0019	0.9940	0.006
20	(Short, Reversal)	0.853	2.906	0.0037	0.9946	0.006
3	(Cash, Long)	0.681	2.601	0.0093	0.9660	0.001
–	DXY buy-and-hold	0.218	0.891	0.373	–	–

$[0.687, 2.194]$. (The first submission reported $[0.490, 1.699]$ and $[0.774, 2.061]$ from an earlier implementation whose seed was not recorded; the values above are produced by the released code with the settings of Table A3.)

8.3. Correcting for the Search

The tests above condition on having chosen the strategy in advance. We did not. We selected the maximum of a 24-strategy set, and the appropriate correction is the Deflated Sharpe Ratio of Bailey and López de Prado [4], which asks whether an observed Sharpe ratio exceeds what the maximum of N trials would be expected to produce under a null of zero true skill, given the observed dispersion of Sharpe ratios across those trials.

The dispersion across our set is very large. Sharpe ratios range from -1.432 to $+1.432$, which pushes the expected maximum under the null to 1.486 in annualized terms. That is higher than Strategy 7's observed 1.064. The resulting Deflated Sharpe Ratio is 0.046. This is the estimated probability that Strategy 7's true Sharpe ratio exceeds the expected maximum of 24 unskilled trials, and it falls far short of the 0.95 required for significance. (The first submission wrongly compared it with 0.05, as if it were a p-value; the conclusion is unchanged but the interpretation is now consistent with Tables 23 and 25.) We therefore cannot reject the hypothesis that Strategy 7's performance is what selecting the best of 24 dispersed trials would produce by chance alone. White's Reality Check [1], with all 24 strategies entering the bootstrap maximum, buy-and-hold as the benchmark, the mean daily return differential as the performance measure and a stationary bootstrap with mean block length 21 and 5,000 replications, gives $p = 0.103$. This is the value reported in Table 16, and across ten further seeds it ranges from 0.095 to 0.113. The first submission reported $p = 0.062$ at this point. That value came from an earlier implementation that used a shorter mean block length, and the Reality Check is sensitive to that choice: with the settings above and the code in `naive_stats.py`, mean block lengths of 5, 10, 21, and 63 sessions give 0.060, 0.072, 0.105, and 0.157. (The 0.103 in Table 16 comes from `exchange_battery.py`, which uses the same settings but draws its bootstrap samples in a different order.) Against a zero benchmark, the Reality Check rejects at $p = 0.008$. The full settings are listed in Table A3.

8.4. Interpretation

Two considerations cut against the pessimistic reading and one reinforces it.

The Deflated Sharpe Ratio assumes N independent trials. Our set is not independent. As Section 6.3 shows, the effective number of trials is between 3.7 and 17.2, depending on

the estimator, and under the smaller estimates the expected maximum under the null falls to 0.75–1.04. Furthermore, the t-statistics in Table 12 are computed on the univariate legs identified in Section 6 rather than on a pattern discovered by scanning. At the time we argued that Strategy 7 was "not found by search" because it is the arithmetic composition of two individually significant legs. That argument is itself a post hoc rationalization: the two legs were identified by looking at the same data, so they are not hypotheses specified in advance.

Against this, Strategy 2 has a Deflated Sharpe Ratio of 0.405, far above Strategy 7's, precisely because its Sharpe ratio of 1.432 sits closer to the null's expected maximum. If one insists on a single result from this study that clears formal multiple-testing correction, it is the overnight leg alone rather than the headline combination. At the smaller effective-trial estimates in Table 7, its DSR rises to 0.958–0.999; at the larger ones it is 0.405–0.789, so this reading depends on the choice of N.

At this point in the analysis, run on the free-provider series alone, the weakest defensible conclusion appeared to be that the sub-period asymmetry in DXY was statistically robust even if the specific 218.8% figure attached to Strategy 7 was not. That conclusion turns out to be wrong, and Section 9 shows why: it is not that the asymmetry survives on weaker grounds than the headline number, but that the asymmetry itself is not present in the independently sourced futures comparator on the same instrument over the same sessions. Every diagnostic in this section is doing exactly what it is designed to do; none of them is designed to ask whether the overnight leg's Sharpe ratio of -1.432 reflects real prices.

9. A Data Defect That the Diagnostics Rank Above the Comparator

9.1. The result and the first attempt to explain it

Decomposing the daily return into an overnight leg from the previous close to the open and a daytime leg from the open to the close produces, on the free series, an overnight leg with a Sharpe ratio of -1.36 and $p < 0.0001$. This is a large effect, and it is essentially the same effect reported as Strategy 1's overnight leg in Section 4 (Sharpe -1.432 on the full 3,932-session sample; the small difference in magnitude is sample overlap, since the comparison here is restricted to the 3,793 sessions common to both series). On the exchange futures for the same instrument, the overnight leg has a Sharpe ratio of $+0.25$ with $p = 0.328$. Both figures are computed on the 3,793 overnight returns available in the 3,794 sessions for which both series report a complete bar, so the two are directly comparable.

A naive way to locate the source is to substitute one vendor's opening prices into the other vendor's series. That test is invalid here. The two series carry a persistent level offset of 0.041 index points, which is 6.7 percent of a typical daily range, so a price level substitution injects the offset into every return. Performing it (the free series' closes with the exchange opens) produced a terminal value of 1,179 for Strategy 7, which is not a finding but an artifact of the offset.

9.2. Isolating the open

The valid version transfers the close to open *return* rather than the price, so that each row uses one vendor's price level throughout, and the result is free of the offset. Table 13 reports it.

Table 13. Scale-free decomposition. Each row uses a single vendor's price level throughout and transfers only the close-to-open return, so the level offset between vendors cannot contribute. The daytime leg is the residual, which makes the 24-hour return identical across rows by construction. Terminal values start from 100, and the sample is the 3,793 overnight returns that are common to both series.

	Overnight leg			Daytime leg		
Price level, opening return	**Final**	**Sharpe**	p	**Final**	**Sharpe**	p
Free, free open	61.68	−1.358	0.0000	197.35	0.700	0.0086
Free, exchange open	108.08	+0.252	0.3280	112.62	0.151	0.5534
Exchange, exchange open	108.08	+0.252	0.3280	112.22	0.146	0.5668
Exchange, free open	61.68	−1.358	0.0000	196.65	0.678	0.0103

The overnight leg follows whichever vendor's open is used and is unchanged by whose closes are used. This is as close to a controlled experiment as observational data allows. It shows that the opening prices are the source of the difference between the two series. It does not by itself show why the free series' opens differ; that question is taken up in Section 9.3.

Table 14 shows what is wrong with them. With the level offset removed, the free series' open lies outside the exchange's high to low range on 7.30 percent of sessions. On the 204 sessions where the free open equals its own daily low, it sits below the exchange low, at −0.073 of the exchange range, and falls outside that range on 56.9 per cent of those sessions against 3.5 per cent on ordinary sessions. A price outside the futures session range is not a price at which the futures contract traded, although because the two instruments differ in construction and session timing a small share of such observations would be expected even for clean data. The free series also records an open exactly equal to the previous close on 6.84 percent of sessions against 1.34 per cent for the exchange series, a pattern typical of stale prints. Section 9.3 shows, however, that stale prints alone cannot produce the overnight drift.

Table 14. Location of the recorded open within the exchange session range, with the level offset removed. A value of 0 is the exchange low and 1 the exchange high. The sessions on which the free open equals its own extreme are the sessions that carry the anomaly.

Session group	n	**Open location**	**Outside range**
Free open at its own low	204	−0.073	56.9%
Free open at its own high	82	+0.994	45.1%
All other sessions	3,508	+0.494	3.5%
Exchange open, all sessions	3,889	+0.498	0.00%

9.3. What the comparison establishes about the mechanism

The free provider does not publish its bar construction, so the mechanism cannot be observed directly. We can, however, narrow down which explanations are consistent with the data by comparing the two sources field by field and session by session. Table 15 summarises five checks.

Table 15. Mechanism checks on the free-provider opening prices (3,793 common sessions, roll sessions excluded). Offsets are free minus exchange, in index points. "Overnight offset change" is the change in the free–exchange offset from the previous close to the open, and "daytime offset change" from the open to the close. A carry-basis effect would move both in the same direction, while an error specific to the open would move them in opposite directions.

Check	Result
(a) Timing: free legs regressed on exchange legs (HAC t in parentheses)	
Free overnight on exchange overnight / exchange daytime	0.808 (25.5) / 0.014 (2.5)
Free daytime on exchange overnight / exchange daytime	0.127 (3.6) / 0.957 (136.4)
Intercept of the free overnight regression	-1.53 bp per session ($t = -5.81$)
(b) Offset changes, index points per session	
Overnight offset change, full sample	-0.0142 ($t = -8.70$)
Daytime offset change, full sample	$+0.0133$
Sum (carry-basis convergence)	-0.0009
Overnight offset change in each three-year window	-0.031, -0.011, -0.013, -0.007, -0.009
t-statistics	-5.5, -3.4, -4.5, -2.9, -3.1
(c) Stale or displaced? Sessions where the free open equals its own low ($n = 204$)	
Share with free open equal to its own previous close	4.4% (7.3% on other sessions)
Mean signed distance to the exchange open (share of range)	-0.317 (-0.009 on other sessions)
Overnight return on these sessions, free / exchange	-16.7 / $+3.1$ bp
(d) Stale sessions: free open equal to its own previous close ($n = 272$)	
Exchange overnight return on the same sessions	-0.14 bp
(e) Independent spot cross-check: synthetic DXY from the free constituent spot bars	
Free index open minus synthetic open, net of the prior-close gap: at own low/high / other	-41.0 / $+37.2$ / -0.5 bp
Overnight Sharpe ratio, free index / synthetic from constituents	-1.37 / $+0.22$
Daily close changes, correlation with H.10 noon synthetic: free / exchange	0.857 / 0.864

Timing. Panel (a) locates the free open in time relative to the exchange session. The free overnight return loads 0.81 on the exchange overnight return and almost nothing on the exchange daytime return. The free daytime return absorbs about 13% of the exchange overnight move. This suggests that the free bar opens somewhat before the exchange bar, or records an opening value that does not fully reflect the overnight move. A timing difference of this kind shifts variance between the legs, but it cannot create a drift. The drift appears in the intercept: -1.53 basis points per session ($t = -5.81$) that is not explained by any exchange return.

An open-specific bias, not carry and not staleness. Panel (b) separates the two explanations that could produce a persistent difference between a spot index and a futures contract. Convergence of the futures basis towards spot would move the free–exchange offset in the same direction over the overnight and daytime legs. Instead, the offset falls by 0.014 index points from the previous close to the open and recovers by 0.013 from the open to the close, so the two moves net to almost zero. That is the signature of an error in the open alone, as if the recorded open were biased downward by about 1.5 basis points on an average session. The overnight bias is significant in each of the five three-year windows, so the defect is persistent rather than a single bad patch. It is, however, far from stationary in size: it is largest in 2011–2013, and Figure 7 shows that 87 of the 90 extreme-located opens in 2013 sit at the session low. Panels (c) and (d) show that stale prints are not the mechanism. On the sessions that carry the anomaly, the free open is *less* often equal to its own previous close than on other sessions, and it sits about a third of the range below the

exchange open. Conversely, on the 272 sessions where the free open is stale, the exchange overnight return is −0.14 basis points, essentially zero. The defect-injection experiment in Section 9.5 confirms that randomly placed stale opens cannot manufacture a drift. The first submission described the defect as a "stale-print defect". We now describe it more precisely as a persistent downward bias in the recorded open, consistent with stale, misfielded, or otherwise inaccurate opening observations.

Independent spot cross-check. Panel (e) replaces the futures comparator with spot prices. Using the ICE formula [46], we build a synthetic DXY from the free provider's own constituent spot bars, which come from a separate pipeline (they are date-stamped one session later than the index file, and are aligned before use). On the flagged sessions, the free index open lies about 40 basis points below (at its own low) or above (at its own high) the synthetic open, against −0.5 basis points on other sessions. The overnight leg of the synthetic index has a Sharpe ratio of +0.22, close to the futures comparator's +0.25 and far from the free index's −1.37. Two caveats apply. The constituent bars follow their own session convention, so their overnight window does not coincide with the index's, and the daily overnight correlation is therefore low (0.07). The Federal Reserve H.10 noon rates provide one observation per day, so they can confirm the closes, whose daily changes correlate 0.86 with the H.10 synthetic for both sources, but not the opens.

Formal constituent decomposition. The index is a geometric average of its constituents, so its log return is an exact weighted sum of the constituents' log returns: $\ln \mathrm{DXY}_t = \ln 50.1435 - \sum_j w_j \ln X_{j,t}$, with X_j quoted in U.S. dollars per unit of currency j and w_j the published weights. Applying the identity to the overnight leg gives Figure 6. On the exchange-sourced data, the cumulative overnight log return of the index futures (+0.112) is matched by the weighted five currency futures (+0.128; the omitted krona accounts for part of the difference). The daily overnight regression has a slope of 1.04 and an R^2 of 0.66. On the free data the index's cumulative overnight log return is −0.482, while the weighted constituents sum to +0.096, leaving an index-specific residual of −0.579. Most of that residual accumulates in 2013, and the rest builds up steadily afterward.

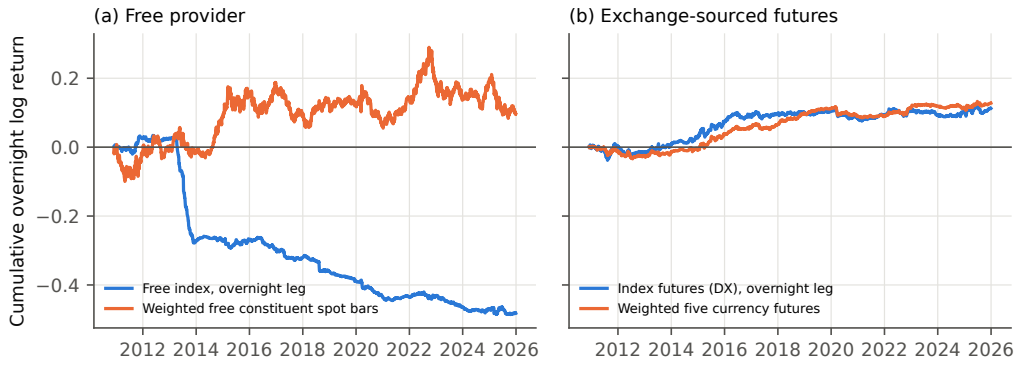


Figure 6. Formal decomposition of the overnight leg using the index's log-linear formula. (a) Free provider: cumulative overnight log return of the index against the weighted sum of the provider's own constituent spot bars. (b) Exchange-sourced futures: index futures against the weighted sum of the five currency futures (krona omitted). The exchange index is accounted for by its constituents; the free index is not.

Robustness to the 2013 cluster. Because the extreme-located cluster opens in 2013, we repeat the main comparison excluding that year. The free overnight leg keeps a Sharpe ratio of −0.69 ($p = 0.010$) on the remaining 3,671 sessions and −0.99 ($p = 0.0008$) over 2014–2025, while the exchange overnight leg is +0.20 ($p = 0.47$) excluding 2013. The free-series overnight Sharpe ratio is negative in every three-year window (−2.46, −0.18, −1.55, −0.96, −1.15) and the exchange ratio is positive in every window.

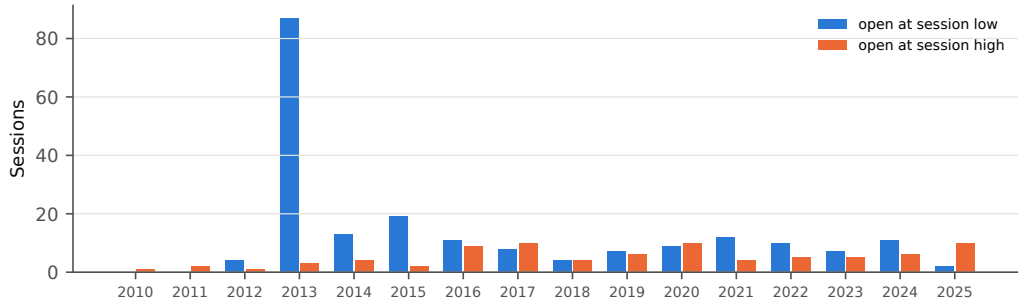


Figure 7. Sessions on which the free-provider open equals the session low or the session high, by calendar year. Under the null of Section 9.5 the two bars in each year are equally likely to be the larger.

What remains unknown. These checks rule out carry convergence, pure staleness, and timing misalignment as sufficient explanations. They also show that an independent spot construction agrees with the futures comparator rather than with the free index. They do not identify the vendor's procedure. A first electronic tick, a quote taken from one side of the market, a misfielded session extreme, or a mixture of these would all be consistent with what we observe. Throughout the rest of the paper, the "defect" therefore refers to this documented, open-specific bias, and its mechanism remains inferred rather than observed.

9.4. What the diagnostics say

The 24-strategy grid of Section 4 (Table 6) was rerun on both series. On exchange data, no strategy beats buy and hold at any cost level, the analysis of variance across the 24 gives $F(23, 93312) = 0.306$ with $p = 0.9994$ and the largest Sharpe ratio in the grid, $+0.269$, is below the $+0.316$ that the null expects the maximum of 24 trials to produce. The walk forward over 13 out-of-sample years ends at 97.69 against buy and hold at 122.64. On the futures comparator, none of the 24 strategies shows evidence of predictability.

Table 16 is the central result of this section. Applied to the free series at zero cost, Hansen's test rejects at $p = 0.013$, three configurations survive the stepwise procedure of Romano and Wolf [3] and the probability of backtest overfitting is 0.016. Applied to the futures comparator the same diagnostics report 1.000, zero survivors and 0.779. The diagnostics rank the defective result far above the comparator's.

Table 16. Multiple testing and overfitting diagnostics applied to the same 24 strategy grid on the two dollar index series, at zero transaction cost. RW is the number of configurations surviving the stepwise familywise error procedure of Romano and Wolf [3]. PBO is the probability of backtest overfitting from the combinatorially symmetric cross validation of Bailey et al. [5], reported at 12 and 16 blocks. Lower PBO indicates a result that persists out of sample. RC and SPA use buy-and-hold as the benchmark, the mean daily return differential of each strategy over it, all 24 strategies in the bootstrap maximum, and a stationary bootstrap with a mean block 21, 5,000 replications and seed 20260828 (Table A3).

Series	White RC p	Hansen SPA p	RW survivors	PBO
Free provider (defective opens)	0.1032	0.0132	3 of 24	0.016 to 0.028
Exchange futures (comparator)	0.9924	1.0000	0 of 24	0.765 to 0.779

Both readings are correct on their own terms. The probability of backtest overfitting asks whether a strategy selected in sample continues to win out of sample. A persistent data defect continues to win, because it is present in both halves of the sample. To every

overfitting diagnostic that conditions on the data, a persistent artifact and a real anomaly are the same object. This is a structural limitation of the entire family of tools and not a defect in any one of them.

9.5. A screen that requires no reference data

The comparison above required a second, independently sourced series. Most analysts do not have one. We therefore propose a screening diagnostic that uses only the series being examined. We present it as a screen, a cheap first check whose rejection calls for investigation, and not as a general test for data corruption.

The null hypothesis. Let P_s, $s \in [0,1]$, be the price path within a session, with $O = P_0$, $H = \max_s P_s$ and $L = \min_s P_s$. The null is not that the open is a random draw from the session range, which the microstructure would not support. It is a symmetry condition:

$$\mathrm{H}_0: \quad \Pr(O = L) = \Pr(O = H), \quad \text{equivalently} \quad \Pr(O = L \mid O \in \{L, H\}) = \tfrac{1}{2}. \tag{5}$$

A sufficient condition is that the law of the path of increments $(P_s - P_0)_s$ is invariant to a change of sign. Reflecting the path about its starting point then maps the event that the open is the session low onto the event that it is the session high [53]. The condition holds for a driftless random walk or Brownian motion, however the steps are distributed, provided they are symmetric. It also holds with fat tails or stochastic volatility, for any number of intraday price changes, and when prices are rounded to a tick grid or printed at the bid or the ask around a latent price, provided the open is not systematically recorded on one side. It is violated, by design, when the recorded open is systematically displaced in one direction relative to the rest of the session. That is exactly the kind of error that would bias the overnight and daytime legs in opposite directions. It is also violated by a within-session drift that is large relative to volatility. For the dollar index the mean daily return is about 1% of the daily standard deviation, and Table 18 shows that this is too small to matter. Price discreteness and opening procedures do affect the *frequency* with which the open coincides with an extreme [38,42]. Under the symmetry condition they do not affect which extreme it coincides with, so the test conditions on the number of extremes and has no free parameter. Counting the two events gives an exact binomial test:

$$n_{\mathrm{low}} \sim \mathrm{Binomial}\Big(n_{\mathrm{low}} + n_{\mathrm{high}},\ \tfrac{1}{2}\Big). \tag{6}$$

Assumptions of the test. Equation (6) needs two assumptions. The first is the symmetry condition (5) in every session. The second is that, given how many sessions have the open at an extreme, whether each of them is at the low or at the high is independent of the others. Variation across sessions in how often the open is at an extreme does not matter, because the test conditions on that count. Dependence does matter. If low-located sessions cluster in time, the sample carries less independent information than its count suggests, and the exact binomial p-value overstates the evidence. We examine this assumption below and report two dependence-robust versions of the test.

Table 17 applies Equation (6) to eight series. It clears the exchange series at $p = 0.788$ and flags the free one at $p = 3.7 \times 10^{-13}$, the value under independence.

Table 17. The open location binomial screen of Equation (6) applied to eight series. Mean position is the average location of the open within the session range, where 0 is the low and 1 is the high. The screen clears the futures comparator and rejects the free-provider index, using only that series' own open, high and low. Binomial p: exact two-sided test, which assumes independent sessions; dependence-robust versions are in Table 19.

Series	At low	At high	Binomial p	Mean position
Dollar index futures, exchange	64	60	0.788	0.499
Dollar index, free provider	204	82	3.68×10^{-13}	0.479
EURUSD, free provider	90	56	0.0061	0.501
GBPUSD, free provider	69	58	0.375	0.504
JPYUSD, free provider	125	103	0.164	0.489
CADUSD, free provider	66	52	0.231	0.506
CHFUSD, free provider	62	44	0.098	0.505
SEKUSD, free provider	44	45	1.000	0.508

Size under microstructure. To check that the symmetry condition survives realistic microstructure, we simulate 400 samples of 3,800 sessions each. In each session, a latent log price follows a random walk of 300 steps, calibrated to the daily volatility of the index futures, and is observed on the futures tick grid of 0.005 index points. Table 18 reports the rejection rate of the screen at nominal levels of 5% and 1%. Under tick discreteness, bid–ask bounce, fat-tailed increments, stochastic volatility, very few intraday steps, and the sample drift, the rejection rate stays between 3% and 6% at the 5% level. The test fails, as intended, only under a drift ten times the sample mean or when the open is recorded systematically below the latent price.

Table 18. Monte Carlo size of the binomial screen (400 simulated samples of 3,800 sessions, 300 intraday steps unless stated, DX tick 0.005). "Share at extreme" is the average share of sessions on which the open equals the high or the low; "Pr(low | extreme)" is the average conditional share at the low.

Data-generating process	Size 5%	Size 1%	Share at extreme	Pr(low | extreme)
Gaussian random walk, continuous prices	0.048	0.003	0.068	0.502
+ tick discreteness	0.045	0.018	0.072	0.502
+ bid–ask bounce (prints at bid or ask)	0.030	0.005	0.072	0.500
+ fat tails (t, 3 degrees of freedom)	0.040	0.010	0.075	0.501
+ stochastic volatility (drawn independently each session)	0.040	0.013	0.074	0.501
+ drift at the sample mean	0.038	0.010	0.072	0.507
20 intraday steps only	0.060	0.010	0.258	0.500
+ drift at ten times the sample mean	0.588	0.363	0.073	0.567
open recorded 0.5 tick below the latent price	1.000	1.000	0.036	1.000
Dependence across sessions (with tick discreteness)				
log volatility AR(1) across sessions, $\phi = 0.98$	0.045	0.010	0.074	0.500
zero-mean drift AR(1), $\phi = 0.9$, s.d. 10% of daily vol.	0.035	0.005	0.073	0.500
zero-mean drift AR(1), $\phi = 0.9$, s.d. 25% of daily vol.	0.040	0.005	0.074	0.499

Dependence between sessions. Table 19 checks the independence assumption directly. Write $d_t = +1$ if the open of session t is at the low, -1 if it is at the high, and 0 otherwise.

In the six futures series the extreme types show no serial dependence: the lag-one autocorrelation of d_t lies between -0.002 and 0.025, and the runs test on the sequence of lows and highs rejects only for the yen ($p = 0.022$). In the free series the flagged sessions cluster strongly (runs test $z = -4.20$; lag-one autocorrelation of d_t 0.149), most visibly in 2013 (Figure 7). The exact p-value of 3.7×10^{-13} is therefore the value under independence and overstates the strength of the evidence. Two dependence-robust versions still flag the free series. A Newey–West test that the mean of d_t is zero gives $p = 7 \times 10^{-7}$, 2×10^{-4} and 0.013 at 8, 21 and 63 lags, and with 2013 excluded the exact test gives $p = 0.008$ and the 21-lag version 0.016. The most conservative version, a sign test that gives each calendar year one vote and so allows any dependence within a year, does not reject: 10 of 15 years have more opens at the low than at the high ($p = 0.30$). We also extended the size study to dependence across sessions (last three rows of Table 18). With volatility that persists across sessions, or a zero-mean within-session drift that persists across sessions, the size at 5% stays between 0.035 and 0.045, and neither process produces serial dependence in the extreme types (mean lag-one autocorrelation of d_t below 0.005). The simulations remain stylised. They do not model opening auctions or the first print after a trading pause, other session-boundary conventions, changes in tick size, or a vendor's processing steps, and the size of the screen under those conditions has not been examined.

Table 19. Dependence between sessions and dependence-robust versions of the screen ($\delta = 0$). Runs: Wald–Wolfowitz test on the time-ordered sequence of lows and highs (negative z: fewer runs, i.e. clustering). NW: Newey–West test that the mean of d_t is zero, at 8, 21 and 63 lags. Years: number of calendar years with more opens at the low than at the high, out of the years with any difference, and the sign-test p-value.

Series	Low/high	Exact p	Runs z (p)	NW 8	NW 21	NW 63	Years (p)
Dollar index, free provider	204/82	3.7×10^{-13}	-4.20 (<0.001)	7×10^{-7}	2×10^{-4}	0.013	10/15 (0.30)
Dollar index futures	64/60	0.788	-0.35 (0.73)	0.714	0.716	0.735	6/13 (1.00)
Euro futures	21/20	1.000	-1.42 (0.16)	0.876	0.880	0.891	6/12 (1.00)
Sterling futures	41/21	0.015	0.92 (0.36)	0.008	0.004	0.005	9/12 (0.15)
Canadian dollar futures	40/50	0.343	0.33 (0.74)	0.292	0.287	0.284	4/10 (0.75)
Yen futures	35/44	0.368	-2.29 (0.022)	0.325	0.345	0.369	4/14 (0.18)
Swiss franc futures	37/44	0.505	-1.17 (0.24)	0.446	0.464	0.488	8/14 (0.79)

Alternative definitions. Exact equality with an extreme depends on how prices are rounded, so we also define the events as the open lying within a fraction δ of the range from the low or the high. Table 20 reports the screen for $\delta \in \{0, 0.01, 0.02, 0.05, 0.10\}$. The free index is flagged at every δ, with p between 1.9×10^{-13} and 1.1×10^{-4}. The index futures are never flagged. As δ grows, the proximity versions lose power, because they include more sessions unaffected by the defect. The five definitions reuse the same sessions and nest one another, so they are not independent tests. That the free index is flagged under all five is therefore a sensitivity check of the definition, not a separate confirmation of the significance level at $\delta = 0$. Treated as a family of five tests, the free index remains flagged after a Bonferroni correction (largest adjusted $p = 5.4 \times 10^{-4}$), but these p-values, like that at $\delta = 0$, assume independent sessions.

Table 20. The screen under alternative definitions of "at the extreme": the open within a fraction δ of the session range from the low or the high ($\delta = 0$ is exact equality). Entries are binomial p-values; the counts are in the replication output.

Series	$\delta = 0$	**0.01**	**0.02**	**0.05**	**0.10**
Dollar index, free provider	3.7×10^{-13}	1.9×10^{-13}	2.3×10^{-10}	2.5×10^{-7}	1.1×10^{-4}
Dollar index futures (DX)	0.788	0.815	0.949	1.000	0.750
Euro futures (E6)	1.000	0.586	0.587	0.496	0.586
Sterling futures (B6)	0.015	0.040	0.455	0.851	0.345
Canadian dollar futures (D6)	0.343	0.783	0.164	0.039	0.049
Yen futures (J6)	0.368	0.714	1.000	0.719	0.244
Swiss franc futures (S6)	0.505	0.923	0.229	0.633	0.549
EURUSD, free provider	0.006	0.011	0.068	0.042	0.144
CHFUSD, free provider	0.098	0.005	0.014	0.759	0.704

False alarms on clean data. The six exchange-sourced futures series are our best available proxy for clean data. At $\delta = 0$ one of the six is flagged at the 5% level (sterling, $p = 0.015$), and none survives a Bonferroni correction across the six. Splitting each series into five non-overlapping three-year windows gives 30 tests, of which 3.3% reject at the 5% level, close to the nominal rate. Across the five definitions in Table 20, four of the 30 futures tests fall below 0.05, against 1.5 expected if the tests were independent (sterling at $\delta = 0$ and 0.01, the Canadian dollar at 0.05 and 0.10). The definitions are strongly dependent, so the definition should be fixed in advance (we recommend $\delta = 0$) rather than chosen after seeing the results.

Power: deliberately injected defects. To study power, we take the clean index futures series and corrupt a random share π of its opens, repairing the high or low whenever a corrupted open falls outside the range, as a vendor's consistency check would. We use five defect types, run 500 injections per cell, and report two quantities: how often the screen rejects at 5%, and how often the corrupted overnight leg becomes "significant" ($|t| > 1.96$). Figure 8 shows the results. They define precisely what the screen can and cannot detect. Writing the session low into the open field is detected with probability 0.998 at $\pi = 1\%$, before it is large enough to manufacture a significant overnight leg (which needs $\pi \geq 2$–3.5%). An open printed outside the range, 71% of the time below it as in the free data, is detected with probability 0.72 at $\pi = 3.5\%$ and 0.94 at $\pi = 5\%$, at which point it produces a significant overnight leg in 12% and 57% of injections. A downward displacement that stays inside the range has low power (0.29 at $\pi = 5\%$) but also produces no significant drift. Stale opens (open equal to the previous close), even when restricted to sessions with an overnight gap up, and symmetric recording noise are never detected. They never manufacture a significant overnight leg either, because they are symmetric or nearly so in their effect on the mean. In these simulations, therefore, the screen detects directional opening-price errors, which are the errors capable of manufacturing an overnight anomaly. It cannot detect symmetric errors, which add variance but not drift. Those are the job of the other checks in the protocol of Section 13, such as the frequency of opens equal to the previous close and cross-vendor comparisons.

Session boundaries and aggregation. The symmetry condition does not depend on where the session boundaries are placed, so the screen can be applied at any aggregation. On weekly bars (Monday open, weekly high and low) the free index is still flagged (25 at the low against 9 at the high, $p = 0.009$) and the index futures are not ($p = 0.23$). Monthly bars contain too few extreme-located opens (6 and 2) to give any power.

What the screen does and does not establish. Three levels of claim should be kept apart. The first is an observed imbalance: in the free series the open equals the session low on 204 sessions and the session high on 82. This is a count and needs no model. The second

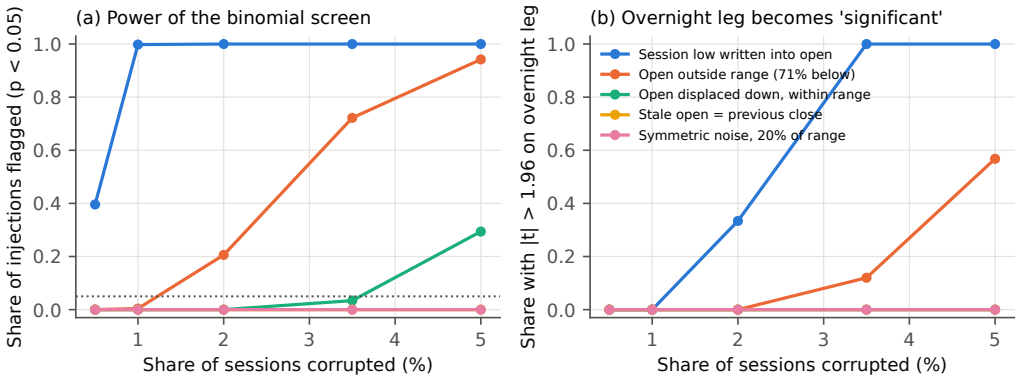


Figure 8. Defects injected into the clean index futures series at controlled rates, 500 injections per point. (a) Share of injections in which the binomial screen rejects at 5% (dotted line: nominal size). (b) Share in which the corrupted overnight leg has $|t_{NW}| > 1.96$. Stale opens and symmetric noise lie at zero in both panels and overlap.

is evidence that the recorded open is biased in one direction. Reading the imbalance that way requires the symmetry condition (5) and an assumption about dependence between sessions, and the strength of the evidence depends on that assumption: $p = 3.7 \times 10^{-13}$ under independence, 7×10^{-7} to 0.013 with the Newey–West versions, and not significant with one vote per year. The directional reading rests as much on the cross-source comparisons of Sections 9.2 and 9.3 (the opening-return swap, the offset that reverses within the day and the synthetic spot index) as on the screen. The third is identification of the process that produced the opens. The screen does not provide it, and neither do the other checks (Section 9.3). The screen's false-alarm rate and power on other instruments and data providers are unknown. They have been examined only on six futures series from one vendor, seven series from one free provider, and the simulated settings of Table 18 and Figure 8.

Two further qualifications. The screen also flags EURUSD at $p = 0.0061$, which barely survives a Bonferroni correction across the eight tests at a threshold of 0.00625. We independently established that these spot series are date-stamped one trading day late, which is consistent with a true positive, but we cannot prove it is not a multiplicity artefact. And the screen is validated here against exactly one instrument with a known good reference, six clean futures series, and the simulations above. It is a screening diagnostic whose rejection calls for investigation. It needs validation on other instruments and vendors before it can be relied on more widely, and it says nothing about errors that are symmetric in direction.

We note also that the mean position of the open within the range at 0.479 for the free series against 0.499 for the exchange series is a weak discriminator. The count of extremes carries the information, not the average location.

10. An Undeclared Search Space

The second failure needs no bad data. A regime conditioned cross-sectional currency strategy, in which six currencies are ranked each quarter, and the two weakest are held with a sign set by the 126-day momentum of the dollar index, reports a Sharpe ratio of 0.72 with a Newey–West t statistic of 2.78 and p =0.0055 on Federal Reserve spot data, once the implementation defects described below are corrected. As originally coded, the same configuration reported 0.79 ($t = 3.09$, $p = 0.0020$).

Before the search space can be assessed, the implementation has to be corrected. The reviewers rightly asked us to keep three kinds of problems apart, because they have different consequences. Implementation defects make the reported numbers wrong for the

specification as stated, and correcting them is not a multiple-testing issue. Specification choices are defensible decisions that should be stated, because the result is sensitive to them. Search decisions determine how many configurations were in fact examined, and they are what the multiple-testing correction must count. Table 21 classifies every problem we found in the committed implementation: four implementation defects, three specification choices and one search decision, each with its correction or justification and its implication.

Table 21. Problems found in the committed implementation of the regime-conditioned strategy, classified by type. Only the search decision affects the multiple-testing denominator. Every result in this section is computed with the four implementation defects corrected. The effect of the third is shown in full in Table 22. "Headline" is the quarterly, bottom-two, 126-session configuration.

Problem	Correction or justification	Implication
Implementation defects		
1. `to_period('2Q')` silently returns quarterly periods	Semiannual periods built explicitly	Semiannual configurations were copies of the quarterly ones, so the grid had five distinct frequencies; the semiannual headline is +0.112 on futures
2. Bare `except` around the grid loop	Removed; every configuration that cannot be built is reported with its reason	Every annual configuration was silently discarded after a `pandas` alias change; all are now evaluated (annual headline +0.456 on futures)
3. `np.where(mom < 0, 1, -1)` maps a missing momentum value to −1	No position until the lookback exists ("Missing signals", below)	The first L sessions were traded as if the dollar were rising; headline 0.79 → 0.72 on spot and 0.54 → 0.46 on futures (Table 22)
4. The free provider's constituent spot files are date-stamped one session late relative to its index file	Not used in this section, which runs on correctly dated futures and Federal Reserve rates	Look-ahead or look-behind in the original regime sign; no result here depends on these files
Specification choices		
5. Benchmark for RC, SPA and DSR was a long dollar position	The book is self-financing long–short, so the null is zero excess return; both nulls reported	Futures RC p is 0.233 against zero and 0.673 against the long dollar (Table 23)
6. Momentum lookback counted on the traded panel's calendar, not the index's own	Original convention kept; the alternative reported	Headline 0.458 versus 0.545 on futures and 0.722 versus 0.867 on spot (replication output)
7. Spot returns exclude the interest differential	Universe C reported both ways	With the differential the spot headline is 0.780 and RC p 0.047; SPA, Romano–Wolf and DSR do not confirm (Table 23)
Search decision		
8. Six configurations reported out of a grid of 216	Full grid declared (Appendix A); every correction uses 144 or 216	The expected maximum under the null, 0.654, exceeds the observed 0.458 on tradeable futures

Missing signals. At each rebalance date the regime sign of a signed configuration is the sign of the dollar index's return over the previous L sessions ($L = 63$, 126 or 252), counted on the traded panel's calendar up to the last session of the formation period. When fewer than L sessions are available, the sign is undefined and the configuration holds no position for the following period. It starts trading at the first rebalance date at which its lookback exists, 1 July 2011 for the quarterly, 126-session headline. Unsigned configurations do not use the regime and are unaffected. The committed code mapped the undefined sign to −1, as though the dollar were rising, so every signed configuration traded from the first rebalance date. Two kinds of statistic are reported in this section and are computed on different samples. Statistics for a single configuration (its Sharpe ratio, t statistic and

p-value) are conditional on its choice and use its own trading sessions. Statistics computed over the grid (RC, SPA, Romano–Wolf, the DSR's $E[\max]$ and the ensembles) use only the sessions on which every configuration in the grid holds a position: 3 January 2012 to 30 December 2025 (3,526 sessions) on the five-currency futures panel, and from 4 January 2012 on the two six-currency panels (3,339 and 3,352 sessions).

Effect of the correction. Table 22 reruns every quantity in this section twice, with the committed mapping and with the correction, and nothing else changed. The correction lowers the headline Sharpe ratio on all three universes, by 0.07 to 0.09, and moves its p-value above 0.05 on both futures panels. It also changes the search-adjusted tests, because signed configurations no longer trade before their lookback exists, which shortens the window on which all configurations trade. With the committed mapping, the Federal Reserve spot universe rejects at the 5% level on White's Reality Check against zero over the full grid ($p = 0.047$), and on both the Reality Check and the SPA test over the 36-configuration subset of Appendix A ($p = 0.015$ and 0.023, with one Romano–Wolf survivor). After the correction none of these rejects ($p = 0.096$; 0.067 and 0.084, no survivor). On the two futures panels no search-adjusted test rejects at the 5% level either before or after. The one-at-a-time sweeps keep their shape. With the committed mapping, the lookback gives -0.319 with no overlay, $+0.177$ at 63, $+0.540$ at 126 and $+0.280$ at 252 sessions, and only the quarterly and annual frequencies are materially positive ($+0.540$ and $+0.413$), as after the correction. The correction therefore removes the rejections that the committed code produced, all on the non-tradeable spot universe. The one that remains, the Reality Check on spot returns with the interest differential added (Table 23), is not confirmed by the other tests. The other conclusions of this section are unchanged.

Table 22. Effect of the missing-signal correction (Table 21, row 3). Every quantity is computed twice, with the committed mapping of a missing momentum value to -1 ("Comm.") and with the correction ("Corr."), and nothing else changed. The headline rows are conditional on the choice of the headline configuration. The remaining rows adjust for the search and use the sessions on which every configuration trades. RC and SPA: stationary bootstrap, mean block 21, seed 20260828, 3,000 replications (5,000 for the subset of 36). The "Corr." columns reproduce Table 23 and the text.

	Futures, 5		Futures plus krona, 6		Federal Reserve spot, 6	
Quantity	**Comm.**	**Corr.**	**Comm.**	**Corr.**	**Comm.**	**Corr.**
Headline Sharpe ratio	0.540	0.458	0.587	0.498	0.792	0.722
Headline Newey–West t (p)	2.11 (0.0349)	1.77 (0.0765)	2.27 (0.0235)	1.90 (0.0570)	3.09 (0.0020)	2.78 (0.0055)
Headline first trading day	2011-01-03	2011-07-01	2011-01-04	2011-07-01	2011-01-04	2011-07-01
Sessions on which all configurations trade	3,778	3,526	3,579	3,339	3,593	3,352
Best of the grid (rank of the headline)	0.632 (2)	0.626 (6)	0.740 (2)	0.669 (5)	0.792 (1)	0.745 (2)
White RC p, null zero	0.2077	0.2333	0.0823	0.2203	0.0470	0.0963
Hansen SPA p, null zero	0.3077	0.3833	0.0907	0.2830	0.0650	0.1570
White RC p, null long dollar	0.6160	0.6730	0.2630	0.4373	0.2457	0.3173
Hansen SPA p, null long dollar	0.7243	0.7647	0.3910	0.5933	0.3753	0.4727
Romano–Wolf survivors	0	0	0	0	0	0
$E[\max]$ under the null; DSR	0.647; 0.337	0.654; 0.227	0.709; 0.321	0.707; 0.217	0.654; 0.701	0.646; 0.612
Ensemble, bottom slice, regime-signed: Sharpe (p)	0.207 (0.42)	0.148 (0.57)	0.397 (0.13)	0.343 (0.21)	0.349 (0.18)	0.312 (0.26)
Ensemble, all regime-signed: Sharpe (p)	-0.103 (0.68)	-0.160 (0.53)	-0.214 (0.40)	-0.265 (0.31)	-0.277 (0.29)	-0.322 (0.23)
Subset of 36: White RC p; SPA p (Romano–Wolf)	0.215; 0.298 (0)	0.450; 0.462 (0)	0.188; 0.209 (0)	0.554; 0.550 (0)	0.015; 0.023 (1)	0.067; 0.084 (0)

Table 23 reports what happens when the search space is declared in full. The original account reported six configurations. Frequency, cross-sectional slice, cardinality, the regime filter, and the momentum lookback are all free parameters and enumerating them gives 144 configurations on the tradeable five-currency panel and 216 on the six currency panels. The 216-configuration grid is the full Cartesian product of six rebalance frequencies (daily, weekly, monthly, quarterly, semiannual, annual), three cross-sectional slices (top, median,

bottom), three cardinalities (one, two, or three currencies) and four regime options (no filter, or a filter with a 63-, 126-, or 252-session lookback). On the five-currency panel, the 72 configurations with cardinality three are infeasible and are excluded, which leaves 144. Every configuration is listed in Appendix A, together with the six that were originally reported.

Table 23. Multiple testing over the declared search space. The strategy is a self-financing long and short book, so the correct null is zero excess return; the committed code tested against a long dollar position, which is shown for comparison. The number after each universe is the count of currencies in it. No configuration survives the stepwise procedure in any universe once the full search is declared. All values are computed with the four implementation defects of Table 21 corrected. The last row adds the three-month interbank differential to the spot returns, which makes the spot universe comparable with the futures universes, whose returns already embed it. RC and SPA: stationary bootstrap, mean block 21, 3,000 replications, seed 20260828. "Sharpe" is the headline configuration's own ratio on its trading sessions and is conditional on its choice; RC, SPA and DSR adjust for the search, and RC and SPA use the sessions on which every configuration trades ("Missing signals").

			Null: zero		**Null: long dollar**		
Universe	**Configs**	**Sharpe**	**RC** p	**SPA** p	**RC** p	**SPA** p	**DSR**
Exchange futures, 5	144	0.458	0.2333	0.3833	0.6730	0.7647	0.227
Futures plus krona, 6	216	0.498	0.2203	0.2830	0.4373	0.5933	0.217
Federal Reserve spot, 6	216	0.722	0.0963	0.1570	0.3173	0.4727	0.612
Federal Reserve spot, total return, 6	216	0.780	0.0470	0.0877	–	–	0.678

The deflated Sharpe ratio is a probability, not a p-value, and must exceed 0.95. It reaches 0.678 at best. The expected maximum Sharpe ratio under the null across 144 configurations is $+0.654$, above the $+0.458$ actually observed on tradeable instruments. With the regime correctly left undefined over its own lookback window, no tradeable universe clears a conventional threshold on any test. The one exception is the non-tradeable Federal Reserve spot universe. Once the interest differential is included, White's Reality Check gives $p = 0.047$, but Hansen's SPA test ($p = 0.088$), the stepwise procedure (no survivors) and the deflated Sharpe ratio (0.678) do not confirm it. That universe uses noon fixing rates that cannot be traded at the fixing, so we regard it as a statement about the spot data rather than as a tradeable result.

The configuration is also a spike rather than a plateau. Holding everything else at the reported values on the tradeable five currency panel and moving one parameter at a time, the regime lookback gives -0.319 with no overlay, $+0.136$ at 63 days, $+0.458$ at 126 days, and $+0.259$ at 252 days. The rebalance frequency gives -0.234 daily, -0.081 weekly, $+0.047$ monthly, $+0.458$ quarterly, $+0.112$ semiannually and $+0.456$ annually. Only two of six frequencies are materially positive, and both neighbors of the reported lookback lose most of the effect, which describes a noisy surface rather than a mechanism.

Finally, an ensemble that holds every configuration in a family at equal weight selects nothing and therefore requires no correction. Averaging the net returns of the 36 regime signed configurations in the bottom slice, which charges each member its own full turnover and so is conservative, gives a Sharpe ratio of 0.148 with $p = 0.57$. Widening the ensemble to all 108 regime signed configurations gives -0.160 with $p = 0.53$.

11. A Correctly Specified Search

11.1. Validating the harness before believing it

Before running any search, we validated the evaluation code itself, because a paper about invalid backtests should demonstrate that its own backtest is valid. Table 24 lists the ten checks. The most important is the first. Weights formed from the same day's realized return, applied without the one-period shift, must produce an absurd Sharpe ratio; if they do not, the shift is broken and nothing downstream means anything.

Table 24. Validation of the evaluation harness. All ten checks pass; the eight rows below group the four turnover and statistic checks into two lines. The lookahead injection is the critical one, because the harness must be demonstrably capable of detecting a strategy that cheats.

Check	Result
Perfect foresight weights applied without the shift	Sharpe +24.34
The same weights applied through the harness	Sharpe −0.11
One extra lag applied to 14 real families	none collapse
Full search on simulated zero drift returns, RC p	0.20, 0.07, 0.69, 0.38
Observed best of N on noise against theory	0.823 against 0.846
Sharpe, Newey–West t, PSR, cost arithmetic	reproduced to machine precision
Buy and hold turnover; daily sign flipper turnover	0.00; exactly 2.0 per session
Winning configuration rebuilt from first principles	maximum difference 0.0e+00

11.2. The search and the noise ceiling

We evaluated 1,312 configurations built from published specifications: time series momentum [54], cross-sectional currency momentum [55], carry [56,57], value [58], moving average rules [59], channel breakouts, open interest growth [60] and volatility scaling [61], each crossed with four rebalance frequencies, the regime overlay at three lookbacks, and with or without volatility targeting. Costs are charged at one basis point per unit of turnover. The search space is the Cartesian product of 41 base signals, four rebalance frequencies, four overlay options (none, or the regime overlay at 63, 126, or 252 sessions) and two volatility-targeting options: $41 \times 4 \times 4 \times 2 = 1{,}312$. No configuration was excluded and all 1,312 were run. Appendix A lists the 41 base signals and their parameters.

The natural benchmark is not zero. It is the Sharpe ratio that this same search attains on returns that contain no predictable structure. We construct that benchmark by resampling the actual returns in blocks using the stationary bootstrap of Politis and Romano [62], which preserves the marginal distributions, the fat tails, and the cross-sectional correlation while destroying the time series structure a signal could exploit, then running the entire search on the resampled panel. We call the resulting maximum the *noise ceiling*. It is not a new statistic. It is a Monte Carlo reality check expressed in the units a practitioner reads.

The first submission averaged the ceiling over six replications, which is far too few to estimate the distribution of a maximum over 1,312 configurations. We have rewritten the search engine so that one replication takes under half a second. The new engine reproduces the original harness exactly on all 1,312 configurations: the largest difference in any daily net return is 0. We now report 2,000 replications for the primary specification and 500 to 1,000 for each sensitivity variant. For each replication, we store the full vector of 1,312 Sharpe ratios, not only its maximum. The resampling procedure is documented in full in Appendix B. The noise ceiling is a benchmark: the performance the declared search achieves when there is nothing to find. Falling below it shows that the observed result is typical of what the search produces on unpredictable returns. On its own, it is not a hypothesis test, and the benchmark depends on the resampling scheme chosen. The corresponding formal test is the Monte Carlo p-value, the share of replications whose best

Sharpe ratio equals or exceeds the observed best. We report both.

Table 25. The correctly specified search against its noise ceiling. The ceiling is the best Sharpe ratio the identical 1,312-configuration search attains on resampled returns. We report its mean, Monte Carlo standard error (s.e.), quantiles with order-statistic 95% confidence intervals in brackets, and the Monte Carlo p-value $(1 + \#\{\text{replications with best} \geq 0.653\})/(1 + B)$. The first-submission values (six replications) were 0.712 for the bootstrap and 0.752 for the Gaussian null, and they are reproduced exactly by the first six replications. The first row is conditional on the selection of the best configuration; every other row adjusts for the search over all 1,312 configurations.

Quantity	**Value**
Best configuration on the real data, net of costs	+0.653
Noise ceiling, stationary bootstrap (mean block 21), $B = 2{,}000$: mean (s.e.)	+0.737 (0.003)
median; 90th percentile	+0.725; +0.887 [0.876, 0.899]
95th percentile; 99th percentile	+0.940 [0.925, 0.958]; +1.047
Monte Carlo p-value of the observed best	0.767
Noise ceiling, correlated Gaussian null, $B = 1{,}000$: mean (s.e.); p	+0.725 (0.003); 0.746
Expected best of 1,312 under the null	+0.865
Configurations exceeding the ceiling	**0 of 1,312**
White Reality Check p	0.659
Hansen SPA p	0.773
Romano–Wolf survivors	0
Deflated Sharpe ratio (needs > 0.95)	0.203
Harvey–Liu adjusted p, Bonferroni and BHY	1.000, 1.000
Probability of backtest overfitting	0.556

Table 26. Sensitivity of the noise ceiling to the resampling scheme. The block length controls how much short-run dependence is preserved.

Resampling scheme	B	**Mean (s.e.)**	**90th pct**	**95th pct**	**MC p of 0.653**
Stationary bootstrap, mean block 5	500	0.747 (0.005)	0.901	0.950	0.786
Stationary bootstrap, mean block 21	2,000	0.737 (0.003)	0.887	0.940	0.767
Stationary bootstrap, mean block 63	500	0.734 (0.005)	0.880	0.931	0.774
Stationary bootstrap, mean block 126	500	0.745 (0.005)	0.887	0.939	0.788
Correlated Gaussian	1,000	0.725 (0.003)	0.863	0.905	0.746

Table 25 gives the result. The best configuration found in fifteen years of exchange data is *below* what the same search finds in resampled data. With 2,000 replications, the ceiling is estimated precisely (Monte Carlo standard error 0.003). The observed best, 0.653, lies at about the 23rd percentile of the null distribution of the maximum (Figure 9a), so 77% of replications produce a better "best" than the real data does. The running mean of the ceiling is 0.736 after 25 replications but moves by up to 0.023 over the next few hundred, and it stays within 0.005 of its final value only after about 300 replications. The first submission's six-replication value of 0.712 was therefore 0.025 below the 2,000-replication mean, an imprecise estimate whose conclusion (the observed best below the ceiling) the larger run confirms. Table 26 and Figure 9b show that the conclusion does not depend on the resampling scheme: the mean ceiling ranges from 0.725 to 0.747 and the Monte Carlo p-value from 0.75 to 0.79 across block lengths from 5 to 126 sessions and a Gaussian null. Configuration by configuration, 45 of the 1,312 real Sharpe ratios exceed the 95th percentile

of their own null distribution, compared with 66 expected if no configuration had any skill. The holdout is consistent: selecting on 2011 to 2018 and evaluating once on 2019 to 2025, the in-sample winner ranks 163rd of 1,312 out of sample, and the top ten in sample configurations average a Sharpe ratio of +0.019.

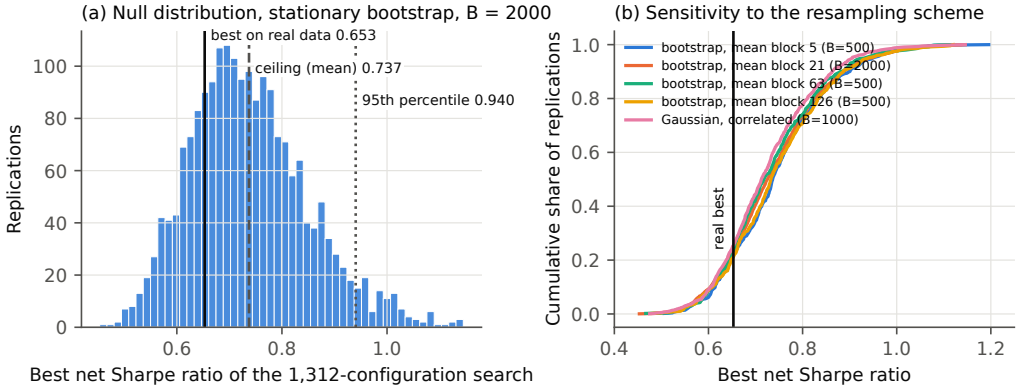


Figure 9. (a) Null distribution of the best net Sharpe ratio of the 1,312-configuration search over 2,000 stationary-bootstrap replications, with the observed best on the real data, the mean (the noise ceiling) and the 95th percentile. (b) Empirical distribution functions of the same maximum under five resampling schemes.

Family ensembles, which select nothing, are reported in Table 27. Not one is significant in either direction.

Table 27. Family ensembles, equal weight within each family, costed in weight space so that trades offsetting across members are not charged. Selecting nothing means no correction is required, so these are a fair test of whether a family of ideas has value.

Family	Sharpe	p	Family	Sharpe	p
Cross sectional momentum	+0.218	0.37	Time series momentum	−0.214	0.37
Value	+0.204	0.43	Volume	−0.232	0.37
Carry	+0.135	0.59	Breakout	−0.289	0.23
Reversal	+0.133	0.58	Moving average	−0.341	0.16
Open interest	+0.094	0.72	Cross sectional reversal	−0.360	0.14
All 1,312 configurations				−0.176	0.48

12. Is Anything Here Profitable?

An exhaustive search that finds nothing invites the objection that the search was the wrong instrument. We therefore also proceeded analytically. Within the five currency futures and the index futures, daily data and the linear measures used here, the structure of the data is clear. The first-order autocorrelation of returns is between −0.004 and +0.033, variance ratios lie between 0.83 and 1.02, and an autoregression of returns on their own lag gives R^2 between 0.0000 and 0.0011. The same autoregression on log realized volatility gives R^2 between 0.966 and 0.972. Volatility is roughly three orders of magnitude more forecastable than direction.

Low linear autocorrelation does not rule out nonlinear, conditional or cross-sectional predictability, so we add three checks (Table 28). (i) The BDS test [63] rejects independence of raw returns for all six series. Once returns are standardized by an exponentially weighted volatility estimate, however, it rejects for none at the 5% level, which points to volatility clustering as the source of the nonlinear dependence. (ii) A state-dependent autoregression allows the lag coefficient to change after large moves and in high-volatility states. It is

jointly significant at the 5% level for one series of six (Swiss franc, $p = 0.026$), and the coefficients are economically small. (iii) An out-of-sample k-nearest neighbor sign forecast based on the last five standardized returns has a hit rate within 2.5 percentage points of 50% for five series. For the yen it is 42.0%, significantly *below* 50% ($p = 0.0008$). This suggests a conditional dependence that the neighbor rule reads with the wrong sign. Exploiting it by inverting the rule would be a new search decision taken after seeing the result, and it would require its own correction and holdout, so we do not pursue it. Cross-sectional predictability is what the momentum, reversal, carry, value, and open-interest families of Section 11 test directly, and none of their ensembles is significant (Table 27). Taken together, these checks do not establish the absence of every form of predictability. They show that the forms we tested are either absent, economically small, or not detectable without a further unadjusted search.

Table 28. Nonlinear and conditional predictability checks on daily futures returns, 2010–2025. BDS: statistics at embedding dimension 2 on raw and on EWMA-standardized returns ($\lambda = 0.94$). State-dependent AR: Wald test that the lag, lag×large-move and lag×high-volatility coefficients are jointly zero (HAC, 8 lags). kNN: share of correct out-of-sample sign forecasts ($k = 50$, five standardized lags, expanding window, 459 forecasts).

Series	BDS raw	p	BDS std.	p	AR Wald p	kNN hit (p)
Euro	7.05	$< 10^{-9}$	−1.86	0.064	0.186	0.505 (0.85)
Sterling	6.71	$< 10^{-9}$	−0.79	0.431	0.203	0.477 (0.35)
Canadian dollar	6.36	$< 10^{-9}$	0.65	0.514	0.060	0.519 (0.46)
Yen	8.02	$< 10^{-9}$	−0.28	0.782	0.892	0.420 (0.001)
Swiss franc	9.93	$< 10^{-9}$	0.25	0.802	0.026	0.525 (0.30)
Dollar index	6.11	$< 10^{-9}$	−1.35	0.178	0.660	0.523 (0.35)

If direction cannot be forecast by these methods then only a genuine risk premium or a relationship true by construction can pay. We tested both. The dollar index is an exact deterministic function of its six constituents, and the index future shares an expiry code with the five currency futures on 98.2 percent of sessions, so a matched synthetic basket is a no-arbitrage relationship. It is also efficiently priced: the convergence trade returns a Sharpe ratio of +0.09 gross and −0.20 at one basis point per leg, because six legs are required against a spread whose daily change has a standard deviation of five basis points.

Carry is the more serious candidate. The roll gap between the expiring and deferred contract is the market's own quoted interest differential, which we validated against the OECD three-month interbank differentials to within 0.43 percentage points per currency. On the fifteen-year futures panel, it returns a Sharpe ratio of 0.377, and the mechanism verifies: of the 1.34 per cent annual gross return, 1.10 per cent is the differential actually held and 0.23 per cent is spot movement, which is the forward premium puzzle behaving as described [64]. Against a noise ceiling built for this five-strategy family with 1,000 replications (the first submission used ten), the observed 0.377 lies below the 90th percentile of the null maximum (0.470) and has a Monte Carlo p-value of 0.21. Table 29 shows why we do not claim it.

Why the two carry samples differ. The first submission's long-sample carry figure, a Sharpe ratio of −0.249, was computed on spot changes alone. It omitted the interest differential, which is the income a carry position is designed to earn, whereas futures returns embed the differential automatically. That figure, and the conclusion the first submission drew from it, are withdrawn. Every long-sample result in this version (Table 29, the factor set below, Figure 10 and the factor-set noise ceiling) uses total returns: the daily spot change plus the three-month interbank differential known at the previous close, accrued at 1/252 of its annual rate per session. Table 29 moves from the long spot sample

to the futures panel one change at a time, so that each step isolates one source of difference. (i) *Return definition* (step 0 to 1): adding the differential moves the long-sample Sharpe ratio from −0.249 to +0.019. Gross of costs, the strategy earns +1.02% a year from the differential and −0.95% from spot movements, +0.08% in total. (ii) *Sample period* (1 to 2): restricting to the futures window gives +0.059. (iii) *Universe* (2 to 3): dropping the krona, which has no liquid futures contract, gives −0.004. (iv) *Instrument* (3 to 4): replacing spot plus differential with within-contract futures returns, keeping the OECD interbank signal, gives +0.128. This step also changes the session calendar and replaces the interbank differential embedded in the returns by the one implied by futures prices, which can differ through deviations from covered interest parity [48]. (v) *Signal* (4 to 5): replacing the monthly OECD interbank differential with the daily roll-implied differential gives the futures panel's +0.377. The split depends on the order of the last two steps. Applying the roll-implied signal to spot total returns first gives +0.318 (step 4′), so the signal accounts for +0.25 to +0.32 of the gap and the instrument for +0.06 to +0.13. Within the common window, most of the difference between the two samples therefore comes from the signal and, to a lesser extent, the instrument. Once returns are defined consistently, the sample period contributes little. The two signals are related but far from identical: the two weight vectors have an average cross-sectional correlation of 0.72, and the spot and futures strategy returns correlate 0.59 to 0.81.

Table 29. Term structure carry: a stepwise reconciliation of the long spot sample with the futures panel (linear cross-sectional carry, one-year smoothed signal, 1 bp per unit of turnover), followed by sub-periods of the long sample on total returns. Each step changes one thing. "Change" is the difference in Sharpe ratio from the previous step, or from step (3) for step (4′). Row (0) is the first-submission construction, which omits the differential; it is shown only to measure the effect of the return definition and is not a result. p is the Newey–West p-value of each specification taken on its own and is not adjusted for any search.

Sample and construction	Change made	Cur.	Sharpe	Change	p
(0) Spot change only, 2002–2026 (withdrawn)	–	6	−0.249	–	0.205
(1) Spot plus interbank differential, 2002–2026	Return definition	6	+0.019	+0.268	0.925
(2) Same, 2010-12 to 2025-12	Sample period	6	+0.059	+0.040	0.817
(3) Same window, krona dropped	Universe	5	−0.004	−0.063	0.987
(4) Futures returns, OECD interbank signal	Instrument	5	+0.128	+0.132	0.620
(5) Futures returns, roll-implied signal	Signal	5	+0.377	+0.249	0.132
(4′) Spot total returns, roll-implied signal	Signal, from (3)	5	+0.318	+0.322	0.200
Sub-periods of the long sample, total return					
2002 to 2007		6	+0.952		0.026
2008 to 2013		6	−0.626		0.106
2014 to 2019		6	−0.070		0.865
2020 to 2026		6	+0.380		0.306

Instability across sub-periods. On total returns, the long-sample carry strategy earns a Sharpe ratio of +0.95 in 2002–2007 and −0.63 in 2008–2013, a pattern consistent with the 2008 carry crash and its aftermath [65,66]. A HAC Wald test of equal means across the four sub-periods gives $p = 0.064$, so equality is not rejected at the 5% level. On the futures panel, which begins in December 2010 and so after the crash, sub-period Sharpe ratios range from +0.18 to +0.84 and are not significantly different ($p = 0.75$). Figure 10 shows the rolling three-year Sharpe ratios of the two constructions. Where they overlap, they agree from about 2020 onward but diverge in 2015–2019, when the spot construction is negative and the futures construction is positive. Two conclusions follow. The swings

across the long sample coincide with the 2008 carry crash, a period the futures panel does not cover, although the differences between sub-periods are not significant at the 5% level. The remaining gap within the common window reflects the signal and the instrument, as the stepwise reconciliation shows.

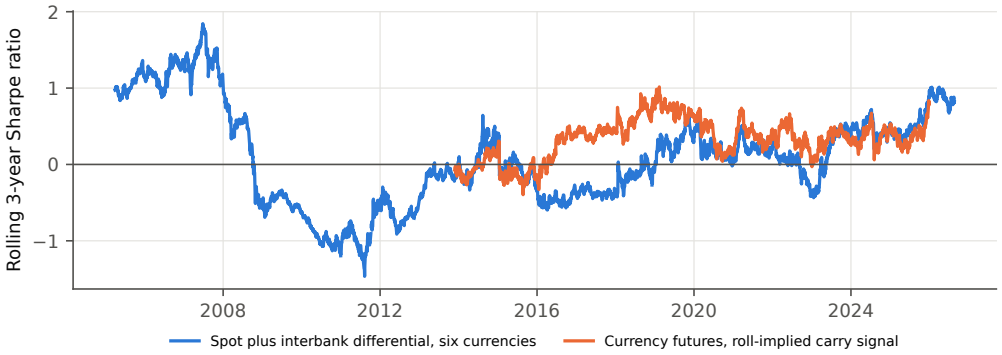


Figure 10. Rolling three-year Sharpe ratio of the linear carry strategy on the long spot sample (total return, six currencies) and on the futures panel (roll-implied signal).

The canonical factor set on the long sample, now on total returns, gives carry $+0.019$, momentum $+0.063$ and value $+0.097$, with an equal risk combination of $+0.122$, against a noise ceiling whose mean is $+0.174$ and whose 90th percentile is $+0.371$ (1,000 replications; Monte Carlo $p = 0.62$). Volatility targeting, the one quantity that is forecastable, changes the Sharpe ratio of a long position by -0.009 on average across seven series over 27 years of price returns and improves three of them. On total returns from 2002 the average change is -0.026, again with three of seven improved. For the dollar index the change (-0.009) is inside the block-bootstrap null. What it does deliver is a reduction in the volatility of realized volatility from 0.025 to 0.015. That is risk control and not return generation.

Scope of the null result. We find no strategy whose performance survives correction for the search *within the scope examined*, and no result here is an estimate of attainable trading profit. That scope is the five CME currency futures and the ICE dollar index futures over December 2010 to December 2025, and Federal Reserve spot rates with OECD interbank rates for six G10 currencies from 2002 to 2026. It covers daily data, the strategy families of Section 11 and this section, rebalancing from daily to quarterly (to annual for the regime strategy), and proportional costs of one basis point per unit of turnover. The reasons are specific to this setting. We find no evidence of linear directional predictability at the daily horizon. The nonlinear checks above find little that survives correction. The one candidate risk premium varies widely across sub-periods of the long sample and is close to zero in total. And five to six highly correlated currencies, with a common factor explaining 54.8 percent of variance, leave little room for a cross-sectional strategy. None of this speaks to other currencies, intraday data, options, other cost assumptions, or other strategy families.

13. A Reporting Protocol

The two failures arise at different stages of the research process, so the protocol is organized by stage. Each stage asks a different question and has its own reporting items (Table 30). The first stage is new relative to existing templates [31,45]. The second and third make existing recommendations operational, and the third adds the noise ceiling.

Table 30. The reporting protocol, by stage. Items marked $*$ would have flagged the first failure, and items marked † the second.

Stage and question	Report
1. Data validation: are the input prices at which the instrument could have traded?	Data source, ticker, retrieval date, and session definition; timestamp alignment against an independent series (lead–lag correlation)*; missing sessions and how they are treated; OHLC consistency violations; share of opens equal to the previous close*; the open-location binomial screen, pre-specified at $\delta = 0$, with a dependence-robust p-value*; for every field, the cross-vendor offset and its overnight and daytime changes, where a second source exists*; contract-roll handling; whether the assumed execution prices (open, close, fix) were tradable.
2. Strategy specification: what was decided before the data were examined?	Hypotheses and the economic rationale were stated in advance, kept separate from explanations offered afterward; the strategy family and every free parameter with its range; the benchmark and the null for each test; implementation defects found and corrected (Table 21 as a template).
3. Multiplicity correction: how unusual is the best result among everything that was tried?	The total number of configurations evaluated, including those discarded and why†; resampling-based tests with full settings (Table A3); an effective number of trials if a DSR is reported; family ensembles; the noise ceiling with the number of replications ($\geq$ 1,000), its Monte Carlo standard error, upper quantiles and the Monte Carlo p-value, and its sensitivity to the resampling scheme†.
4. Economic interpretation: is the surviving result worth having?	Costs from spreads at the execution times, slippage and impact where available, otherwise a break-even cost; capacity; out-of-sample and sub-period stability; and a clear statement of which mechanisms were hypothesized before the test and which were proposed afterward.

Screen the inputs. Report the binomial statistic of Equation (6), with a dependence-robust version such as the Newey–West test of Table 19, the frequency with which the open equals the previous close, and the count of sessions violating the open, high, low, and close ordering. These cost one pass over the data and would have flagged the first failure for investigation before any strategy was run. Because the binomial screen cannot detect symmetric errors (Section 9.5), it complements rather than replaces the other stage-1 checks.

Declare the whole search. Report every parameter that was varied and the total count of configurations evaluated, including those discarded. Report the ensemble across each family alongside the selected configuration, since an ensemble selects nothing and needs no correction.

Report the noise ceiling. Run the complete search on block bootstrapped returns at least 1,000 times and report the distribution of the best Sharpe ratio it produces: its mean, Monte Carlo standard error, upper quantiles, and the share of replications at or above the observed best. A result below the ceiling cannot be distinguished from what this search produces on unpredictable returns under the chosen resampling scheme. This is a benchmark statement, not proof that the result is due to luck. The Monte Carlo p-value supplies the formal test. In this study the ceiling was 0.737 (95th percentile 0.940; 2,000 replications) for the 1,312 configuration search, 0.221 (90th percentile 0.470; 1,000 replications) for a five strategy carry family and 0.174 (90th percentile 0.371) for the canonical factor set. Every real result fell below its own 90th percentile, with Monte Carlo p-values between 0.21 and 0.77.

14. Discussion: A Plausible Story Is Not Evidence

Before the defect in Section 9 was identified, we had a ready set of economic explanations for the overnight leg's negative drift. It is worth recording them, because the fact that they were easy to construct is itself the paper's point. All three were proposed after the result was seen. None was a hypothesis specified in advance and then tested, and that difference is the reason they deserve no evidential weight.

Carry. The DXY basket is dominated by EUR at 57.6% and JPY at 13.6%. Over most of 2010–2025, U.S. policy rates exceeded the euro area and Japanese rates, so a long dollar position accrued positive carry, and a mechanism by which that carry is realized disproportionately outside the daytime session is not implausible on its face. This is consistent with the currency risk premium literature [56,57], which identifies the interest rate differential as the dominant source of expected currency returns.

Liquidity provision. The close-to-open interval spans the thinnest liquidity window for USD-denominated risk. If dealers demand compensation for holding inventory through it, the price at the open should be systematically displaced relative to the prior close in the direction that penalizes the inventory holder. Documented time-of-day patterns in currency returns [35,36] would have made this account look supported by the literature.

Settlement mechanics. A contract with a defined settlement window can exhibit return patterns near that window that have nothing to do with an economic risk premium.

Each of these stories is independently reasonable, each is grounded in real literature, and none of them is why the overnight leg in Section 4 has a Sharpe ratio of -1.432. The evidence in Section 9 points instead, without establishing it conclusively, to a persistent downward bias in the free provider's recorded opening prices, concentrated in 2013 and present in every window. That evidence includes opens lying outside the futures session range, an open-specific offset that reverses within the day, and disagreement with a synthetic index built from the provider's own constituents. No carry mechanism, liquidity story, or settlement-window argument is needed, because the same leg on the independently sourced futures comparator, and on the synthetic spot index, is statistically indistinguishable from zero.

We flag this not to criticize the exercise of proposing mechanisms – it is a normal and useful part of interpreting a backtest – but because the ease with which one can be proposed is exactly what makes a data defect dangerous. A result that survives an economically literate narrative feels more credible than one that does not, and that feeling is uncorrelated with whether the underlying prices are real. The two failures documented in this paper – uninspected inputs and undeclared search spaces – both produce results that are easy to rationalize after the fact. The discipline that catches them has to come from the diagnostics in Sections 9 and 11, not from the plausibility of the story attached to the result. This discipline has limits in our own case. The diagnostics were developed and validated on one instrument complex with two principal data sources, and the mechanism of the defect is inferred rather than observed. How well the screen and the noise ceiling work elsewhere is a question for further work, not a finding of this paper.

15. Limitations

We flag ten, roughly in order of scope.

1. The evidence is one complex currency and two vendors. Nothing here generalizes to free data as a category or to this vendor across other instruments.
2. The mechanism behind the defect is inferred rather than observed: a stale print, a first electronic tick, and a misfielded session extreme are all consistent with what we see, and we cannot inspect the vendor's pipeline. The checks in Section 9.3 rule out some explanations but do not identify the vendor's procedure.

3. The commercial series is itself a redistributor rather than the exchange, and its closing field is a last price whose settlement status we could not confirm. It is a different instrument from the spot index, so it serves as a comparator, not as ground truth.
4. The screen of Section 9.5 is validated against a single instrument with a trusted reference, six clean futures series, and simulations. Its size and power were studied for the data-generating processes and defect types we specified, but it cannot detect symmetric errors, and its false-alarm rate on other vendors and instruments is unknown.
5. The exact binomial p-value of the screen assumes that sessions are independent. In the free series the flagged sessions cluster in time, so that p-value overstates the evidence. The dependence-robust versions in Table 19 are weaker, and the most conservative of them does not reject. Our simulations of dependence across sessions cover only persistent volatility and persistent drift.
6. The noise ceiling of Section 11 is a presentation of an existing idea rather than a new statistic. It is a benchmark that depends on the chosen resampling scheme: the block bootstrap removes time-series structure while keeping the marginal distributions and cross-correlation, and other schemes would produce somewhat different ceilings. The first submission estimated it from only six replications. That estimate was 0.025 below the 2,000-replication value, and its precision was not reported.
7. The absence of a strategy that survives correction in Section 12 is a statement about six daily G10 series over fifteen to twenty-seven years and not about foreign exchange markets in general; emerging market carry, options, and intraday data all lie outside what we can test.
8. Costs are modeled rather than measured, throughout the paper: we deduct a flat proportional cost (zero to two basis points per trade for the 24-strategy grid, one basis point per unit of turnover from Section 10 onward) and do not model bid-ask spread variation, slippage at the open, or market impact. The spread estimates of Appendix D come from daily bars and do not measure these costs.
9. Execution at the open and close is assumed in Sections 4–9, at prices that a systematic participant cannot generally guarantee. Neither this nor the previous limitation changes the paper's conclusion, since the naive result fails for a more basic reason before either one becomes relevant, but both would matter for any future backtest that clears the data and search-space screens proposed in Section 13 and needs to be evaluated on its economics.
10. The Federal Reserve spot universes in Sections 10 and 12 are built from noon fixing rates and monthly interbank rates. They are not tradeable, and the carry reconciliation shows that construction choices of this kind can move a Sharpe ratio by several tenths.

16. Conclusions

Two failures sit outside the reach of the standard defense of a backtest, and this paper documents both on the same instrument. The data show the following. On a widely used free source, a sub-period decomposition of the DXY produced a strategy returning 218.8% with a Sharpe ratio of 1.06. Standard diagnostics rated the result favourably: Hansen's test rejected, and the probability of backtest overfitting was 0.02. On an independently sourced futures comparator for the same sessions, the overnight leg is statistically indistinguishable from zero (Sharpe $+0.25$, $p = 0.328$), no strategy in the same grid beats buy-and-hold, and the same diagnostics rate the result as indistinguishable from noise. The evidence is consistent with a persistent, open-specific downward bias in the free series rather than with a market effect: an offset that reverses within the day, opens outside the futures range, and disagreement with a synthetic index built from the provider's own constituent

prices. It supports this interpretation but does not establish it conclusively, and it does not identify the process that produced the opens. Because such a defect persists out of sample, overfitting diagnostics that condition on the data rank it above the comparator. A binomial screen on the location of the recorded open flags the free series without any reference data. Its exact p-value assumes independent sessions; allowing for the clustering of the flagged sessions the series is still flagged at the 5% level by the Newey–West versions, but the evidence is weaker than the exact test suggests. In the simulated settings we examined, it holds its nominal size under symmetric microstructure and detects directional opening errors before they are large enough to manufacture a significant anomaly, but it cannot detect symmetric errors.

The second failure needs no bad data. A regime-conditioned currency strategy reporting $t = 2.78$ on six configurations ($t = 3.09$ before an implementation defect was corrected) does not survive correction on tradeable instruments once its 216-configuration search is declared. A fully declared search of 1,312 configurations built from published specifications finds nothing above what the same search attains on resampled returns: the observed best lies at the 23rd percentile of a null distribution estimated from 2,000 replications, and between the 21st and 25th percentiles under every resampling scheme we tried. An undeclared search space applies the correction to the wrong denominator, and no amount of care in applying the correction itself fixes that.

These are statements about statistical evidence, not about tradable profit. The costs of one and two basis points used throughout are sensitivity assumptions, and the spread estimates from daily bars do not measure the spreads, slippage, market impact or attainable execution prices of the strategies studied. A claim that any strategy here is profitable would require execution assumptions supported by time-stamped quotes and trades at the execution times.

Several things need validation elsewhere: whether the screen generalizes to other instruments and vendors, and what its false-alarm rate is there; what produced the defect; and whether the null result extends beyond the instruments, data, frequencies, strategy families, and periods examined. The protocol we propose separates data validation, strategy specification, multiplicity correction, and economic interpretation. Following it would have prompted an investigation of the first failure before any strategy was run, and would have exposed the second as soon as the search was counted. We report both failures because a paper that only demonstrated the successful correction of a multiple-testing problem would have missed the two failures that mattered most in our own analysis.

Author Contributions: Conceptualization: E.P. and A.M.; Methodology: A.M.; Software: A.M.; Validation: A.M. and E.P.; Formal analysis: A.M.; Investigation: A.M.; Data curation: A.M.; Writing—original draft preparation: A.M.; Writing—review and editing: A.M. and E.P.; Visualization: A.M.; Supervision: E.P. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: Data and code are available at https://github.com/RyanSingh0/dxy-fx-strategies (accessed on 15 October 2026). `RyanSingh0` is the personal GitHub account of the first author, Aryan Meena, and every commit in the repository is authored under his name. Release `v2.1-revision2` reproduces every table and figure in this version of the paper; release `v2.0-revision1` corresponds to the first revision. All input data are included in the repository. Appendix C gives the computational environment, run instructions, random seeds, the sources of the data and how to obtain them, and the map from each table and figure to the script and output file that produce it.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

ANOVA	Analysis of variance
BDS	Brock–Dechert–Scheinkman test
CAGR	Compound annual growth rate
CC	Close-to-close
CME	Chicago Mercantile Exchange
CO	Close-to-open
DSR	Deflated Sharpe Ratio
DXY	U.S. Dollar Index
FRED	Federal Reserve Economic Data
HAC	Heteroskedasticity and autocorrelation consistent
MC	Monte Carlo
OC	Open-to-close
OHLC	Open, high, low, close
PBO	Probability of backtest overfitting
PSR	Probabilistic Sharpe Ratio
RC	Reality Check (White's Reality Check)
RW	Romano–Wolf stepwise procedure
SPA	Superior predictive ability (Hansen's test)
SR	Sharpe ratio

Appendix A. Enumeration of the Search Spaces

The regime-conditioned strategy (Section 10). Each configuration is a tuple (frequency, slice, cardinality, regime). At each rebalance date the currencies are ranked by their return over the previous period; the strategy holds the k currencies in the chosen slice for the next period, with a sign set by the dollar regime when the filter is on. Table A1 lists the grid. The 216 configurations are its full Cartesian product. On the five-currency futures panel the 72 configurations with $k = 3$ are infeasible: a top or bottom slice of three needs six currencies, and the median slice of three is dropped by the same feasibility rule. The remaining 144 are evaluated. No configuration fails for any other reason, and every dropped configuration is reported with its reason by `build_grid`. The headline configuration of the original account is quarterly, bottom slice, $k = 2$, regime 126.

Table A1. The declared grid of the regime-conditioned strategy.

Parameter	Values	Count
Rebalance frequency	daily, weekly, monthly, quarterly, semiannual, annual	6
Cross-sectional slice	top, median, bottom (by prior-period return)	3
Cardinality k	1, 2, 3	3
Regime filter	none; sign of 63-, 126- or 252-session DXY momentum	4
Total, six-currency universes (B, C)		216
Infeasible with five currencies ($k = 3$): $6 \times 3 \times 1 \times 4$		−72
Total, five-currency futures (A)		144
Author-favourable subset (lookback 126, $k = 2$ fixed): $6 \times 3 \times 2$		36

The author-favourable subset in the last row fixes the lookback and cardinality at their reported values and searches only frequency, slice and the regime switch. Even on this subset the Reality Check gives $p = 0.45$ on the tradeable five-currency panel.

The 1,312-configuration search (Section 11). Each configuration is a tuple (base signal, frequency, overlay, volatility target). Table A2 lists the 41 base signals. Each is crossed with

four rebalance frequencies (daily, weekly, monthly, quarterly), four overlays (none, or the regime overlay with a 63-, 126- or 252-session lookback) and two volatility-targeting options (none, or a 10% target on 63-session realised volatility, capped at three times leverage): $41 \times 4 \times 4 \times 2 = 1{,}312$. Every signal uses only information available at the close of the formation date, and a single one-period shift in the harness applies it. Gross exposure is normalised to one for every family except the volatility-targeted variants. All 1,312 configurations built successfully and enter every correction.

Table A2. The 41 base signals of the 1,312-configuration search.

Family	Specification and parameters	Signals
Time-series momentum	sign of L-session return, $L \in \{21, 63, 126, 252\}$ [54]	4
Time-series reversal	minus the sign of L-session return, same L	4
Cross-sectional momentum	long top k, short bottom k by L-session return, $k \in \{1, 2\}$ [55]	8
Cross-sectional reversal	the opposite of the above	8
Moving-average crossover	sign of fast minus slow mean, $(5, 20)$, $(10, 50)$, $(20, 100)$, $(50, 200)$ [59]	4
Channel breakout	Donchian $L \in \{20, 50, 100\}$	3
Carry	long top k, short bottom k by interbank differential, $k \in \{1, 2\}$ [56,57]	2
Value	five-year reversal, $k \in \{1, 2\}$ [58]	2
Open interest	open-interest growth over $L \in \{21, 63\}$, $k \in \{1, 2\}$ [60]	4
Volume	21-session over 84-session mean volume, $k \in \{1, 2\}$	2
Total base signals		41

Appendix B. Resampling and Multiple-Testing Implementation

All resampling uses the stationary bootstrap of Politis and Romano [62]. A replicate index is built by starting at a uniformly drawn session and, at each step, either moving to the next session (wrapping at the end of the sample) or, with probability $1/\bar{b}$, jumping to a new uniformly drawn session. Block lengths are therefore geometric with mean $\bar{b}$. When several return series are resampled, the same index is applied to every column, which preserves contemporaneous cross-correlation. Our default mean block length of 21 sessions (one trading month) is a conventional choice for daily returns with weak serial dependence. Automatic selection [67,68] would give shorter blocks, because the autocorrelation of returns is close to zero. We report sensitivity to $\bar{b} \in \{5, 10, 21, 63, 126\}$. On the number of replications we follow Davidson and MacKinnon [69] and Andrews and Buchinsky [70]: 5,000 for p-values near conventional thresholds (RC, SPA, Romano–Wolf on the 24-strategy grid), 3,000 on the 216-configuration grid, and 2,000 for the noise ceiling. For the ceiling, the Monte Carlo standard errors are reported directly. Table A3 lists every setting.

Table A3. Implementation settings of every resampling-based procedure. "Differential": the daily series whose mean is tested. RC and SPA statistics are the maximum (studentised, for SPA, with Hansen's recentring) of $\sqrt{n}$ times the mean differential over all strategies listed; Romano–Wolf uses studentised statistics with stepwise 95% critical values from the same bootstrap draws. The last three rows give the seeds of the simulation studies of the screen.

Result	Strategies; differential	Benchmark	$\bar{b}$	B	Seed	Script
Sharpe CI (Section 8)	one strategy; Sharpe ratio	–	21	5,000	1	`naive_stats.py`
RC/SPA, 24 grid, free	all 24; strategy minus benchmark	buy-and-hold	21	5,000	20260828	`exchange_battery.py`
RC/SPA/RW (Table 16)	all 24; as above, both series	buy-and-hold	21	5,000	20260828	`exchange_battery.py`
RC sensitivity (Section 8)	all 24; as above, also vs zero	B&H / zero	5–63	5,000	1–10	`naive_stats.py`
PBO (Table 16)	all 24; CSCV, $S \in \{12, 16\}$ blocks	–	–	all splits	–	`exchange_battery.py`
Table 23	144 or 216; book return	zero / long dollar	21	3,000	20260828	`run_battery.py`
Table 25, RC/SPA/RW	all 1,312; net return	zero / DX	21	2,000	11	`run_search.py`
Noise ceiling, bootstrap	all 1,312; best net Sharpe	–	5–126	500–2,000	$9000 + b^a$	`run_ceiling.py`
Noise ceiling, Gaussian	all 1,312; best net Sharpe	–	–	1,000	$5000 + b$	`run_ceiling.py`
Carry and factor ceilings	5 candidates; best net Sharpe	–	21	1,000	$300/400/880 + b$	`carry_instability.py`
Table 22	144 or 216, both mappings; book return	zero / long dollar	21	3,000; 5,000	20260828	`npwhere_effect.py`
Simulation studies						
Table 18, rows 1–9	simulated samples of 3,800 sessions	–	–	400	20261006	`screen_study.py`
Figure 8	defects injected into DX	–	–	500	7	`screen_study.py`
Table 18, rows 10–12	simulated samples, dependent sessions	–	–	400	20261015	`screen_dependence.py`

[a] For $\bar{b} = 21$; for other block lengths the seed is $9000 + 10^5\bar{b} + b$. b indexes the replication.

In the noise-ceiling replications the resampled pair returns replace the actual returns. Closes are rebuilt by compounding the resampled returns, and highs and lows are set to $\pm 0.2\%$ of the close (used only by the breakout family). The dollar index is replaced by the negative of the equal-weighted resampled pair return, so the regime overlay cannot draw on the real index. Interest rates, open interest and volume are left unchanged, so the carry, open-interest and volume signals keep their real values but are paired with returns whose time ordering has been destroyed.

Appendix C. Reproducibility

Repository and version. The code and data are at https://github.com/RyanSingh0/dxy-fx-strategies, the personal GitHub account of the first author, Aryan Meena. Release `v2.1-revision2` produces every table and figure in this version. Relative to `v2.0-revision1` (the first revision) it adds `src/revision/npwhere_effect.py` (Table 22), `screen_dependence.py` (Table 19 and the last rows of Table 18) and `naive_tables.py` (Tables 2–6 and 8–11). It adds the alternative ordering to `carry_instability.py`, moves the long-sample carry and factor scripts in `src/part4_reasoned/` to total returns so that their output agrees with Section 12, adds a total-return check to `volatility_targeting.py`, and extends `price_swap.py` and `text_numbers.py` to print values quoted in the text. The notebooks in `original/` are the first implementation, kept so that the defects of Table 21 can be inspected; no result in the paper depends on them.

Environment. `requirements.txt` pins the versions used: Python 3.11, `numpy` 2.4.4, `pandas` 3.0.2, `scipy` 1.17.1, `statsmodels` 0.15.0 and `matplotlib` 3.10.9 (`results/revision/environment.txt`). For this version every script was run again from the raw data with these versions under Python 3.11 on a two-core Linux machine. The output of every script not changed in this version was reproduced, including the full matrices of the noise-ceiling replications. All random numbers come from `numpy.random.default_rng` with the seeds in Table A3. `run_all.sh` runs every script in order and writes all output to `results/`; it takes two to three hours on two cores, most of it the noise-ceiling replications. Each script can also be run on its own from `src/`. The fast search engine reproduces the original harness exactly (`fast_search.py` checks all 1,312

configurations, maximum difference 0).

Data and how to obtain them. All inputs are in `data/`. (i) The free-provider series, as downloaded: the dollar index and the provider's six constituent spot bars, daily, December 2010 to December 2025. (ii) The exchange-sourced futures: nearby continuous daily bars with open, high, low, last price, volume and open interest for DX, E6, B6, D6, J6 and S6, downloaded from Barchart on 27 August 2026. They are redistributed only so that the results can be reproduced. A reader who cannot use them under the vendor's terms can obtain the same nearby series from any exchange-sourced vendor and rerun every script; `src/data.py` documents the expected layout and the roll handling. (iii) Federal Reserve H.10 spot rates and OECD three-month interbank rates, retrieved from FRED; `data/README.md` lists the series codes and gives a script that rebuilds the long file from the public API. The free-series analyses (Sections 4–8), the Federal Reserve spot universe of Section 10 and the long-sample results of Section 12 need only public data.

Data processing. The futures files are nearby continuous series with the contract symbol on every row. Returns are computed within contract, and roll sessions are set to zero (Section 3). The free-provider file stores the index and constituents in one table with missing index rows left as missing. Federal Reserve series are used as published, and monthly OECD rates are forward-filled to daily. No outlier filter is applied to any series. This is deliberate: a filter tuned on the free series would have hidden the defect that the paper documents.

Map from results to code. Table A4 lists, for every table and figure, the script that produces it and the file to which it writes. All paths are relative to the repository root.

Table A4. Map from each table and figure to the script that produces it and its output file (scripts in `src/`, output in `results/`).

Result	Script	Output
Tables 2–6, 8–11; break-even costs (Figure 4)	`revision/naive_tables.py`	`revision/naive_tables.txt`
Tables 7, 12; Section 8 bootstrap intervals and Reality Check sensitivity; 2013 robustness	`revision/naive_stats.py`	`revision/naive_stats.txt`, `.json`
Section 8 ANOVA and assumption tests; Newey–West lags; Table 21 rows 3 and 6	`revision/text_numbers.py`	`revision/text_numbers.txt`
Figures 1–5	`revision/figures_naive.py`	`revision/figures/`
Section 3 checks on the futures (volume, dating, rolls)	`part1_data_defect/diagnose_futures.py`	`part1_futures_diagnostics.txt`
Tables 13, 14; level offset and price-level swap	`part1_data_defect/price_swap.py`	`part1_price_swap.txt`
Section 9 frequency of opens equal to the previous close	`part1_data_defect/adversarial_checks.py`	`part1_adversarial.txt`
Tables 16, 17; Section 9 grid on both series	`part1_data_defect/exchange_battery.py`	`part1_exchange_battery.txt`
Table 15; Figures 6, 7	`revision/mechanism.py`, `make_figures.py`	`revision/mechanism.txt`, `figures/`
Tables 18 (first nine rows), 20; Figure 8; false alarms; weekly bars	`revision/screen_study.py`	`revision/screen_study.txt`
Table 19; Table 18 (last three rows)	`revision/screen_dependence.py`	`revision/screen_dependence.txt`
Tables 21, 23; Section 10 sweeps	`part2_declared_search/run_battery.py`	`part2_declared_search.txt`
Table 23, total-return row	`revision/part2_spot_tr.py`	`revision/part2_spot_tr.txt`
Table 22	`revision/npwhere_effect.py`	`revision/npwhere_effect.txt`
Section 10 ensembles; Appendix A subset of 36	`part2_declared_search/ensembles.py`	`part2_ensembles.txt`
Table 24	`part3_exhaustive/validate_harness.py`	`part3_harness_validation.txt`
Table 25 (search and tests), holdout	`part3_exhaustive/run_search.py`	`part3_search.txt`
Table 27	`part3_exhaustive/verify_search.py`	`part3_verification.txt`
Tables 25 (ceiling), 26; Figure 9	`revision/fast_search.py`, `run_ceiling.py`, `summarise_ceiling.py`, `make_figures.py`	`revision/ceiling_summary.txt`, `ceiling_*.npy`, `maxdist_*.npy`
Section 12 autocorrelation, variance ratios, basket trade	`part4_reasoned/analyse_structure.py`, `carry_futures.py`	`part4_structure.txt`, `part4_carry_futures.txt`
Table 28; Table A5	`revision/nonlinear_spreads.py`	`revision/nonlinear_spreads.txt`
Table 29; Figure 10; carry and factor ceilings	`revision/carry_instability.py`, `make_figures.py`	`revision/carry_instability.txt`, `.json`
Section 12 factor set and volatility targeting	`part4_reasoned/factors_long_sample.py`, `volatility_targeting.py`	`part4_factors_long_sample.txt`, `part4_volatility_targeting.txt`

Appendix D. Spread Estimates from Daily Bars

Intraday quotes for the execution times are not available to us, so Table A5 reports three standard estimators of the effective spread from daily bars: Roll [39], Corwin and Schultz [71], and Abdi and Ranaldo [72]. Each is computed by month, with negative estimates set to zero, and the median across months is reported. For instruments as liquid as these, daily-bar estimators are known to be unreliable. They are dominated by volatility over the nearly 24-hour session rather than by the quoted spread, and they disagree with each other by an order of magnitude. We therefore read them only as an indication that one to two basis points per trade is not conservative for a strategy trading at the open and close. Credible all-in costs would require time-stamped quotes and trades at the execution times, an estimate of impact for the intended size, and a check that the recorded open and close were tradable prices. The last check fails for the free series by construction.

Table A5. Effective-spread estimates from daily bars (basis points, median of monthly estimates, roll sessions excluded).

Contract	Abdi–Ranaldo	Roll	Corwin–Schultz
Euro (E6)	1.0	0.0	6.0
Sterling (B6)	7.7	8.5	5.6
Canadian dollar (D6)	5.5	0.0	4.9
Yen (J6)	1.7	15.0	4.6
Swiss franc (S6)	0.0	6.6	6.3
Dollar index (DX)	0.0	0.0	2.2

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